

Internet Appendix for

“Economic Fundamentals and Short-Run Exchange Rate Prediction: A Machine-Learning Perspective”

August 19, 2026

IA.1. Data

This section provides details on the data sources and construction of the country characteristics and global variables.

A. Exchange Rates

Daily bid and ask spot and forward exchange rates for 14 countries are from Barclays and Reuters via Datastream. Datastream country mnemonics for the currencies are as follows:

Country	Mnemonic	Country	Mnemonic
United Kingdom	GBP	Norway	NOK
Switzerland	CHF	Denmark	DKK
Japan	JPY	Euro area	EUR
Canada	CAD	Germany	DEM
Australia	AUD	Italy	ITL
New Zealand	NZD	France	FRF
Sweden	SEK	Netherlands	NLG

Bid spot price Ticker BB***SP(EB), where “***” indicates the country mnemonic.

Ask spot price Ticker BB***SP(EO).

Bid forward price Ticker BB***1F(EB).

Ask forward price Ticker BB***1F(EO).

We use the midpoint of end-of-month bid and ask spot prices to compute the month- t spot exchange rate for country i , which we denote by $S_{i,t}$ and is expressed as the number of country- i currency units per US dollar. We then calculate the country- i log exchange rate change:

$\Delta s_{i,t} = s_{i,t} - s_{i,t-1}$, where $s_{i,t} = \log(S_{i,t})$. The set of exchange rates is similar to the sample of developed countries employed in other studies (e.g., Lustig, Roussanov, and Verdelhan (2011), Menkhoff, Sarno, Schmeling, and Schrimpf (2012a)), except Belgium, which we exclude due to a lack of data availability for the country characteristics. Table IA.1 reports summary statistics for the log exchange rate changes for the 14 countries.

B. Country Characteristics

Country characteristics are computed using data from Finaeon and the Organization for Economic Cooperation and Development (OECD). Finaeon and OECD country mnemonics are as follows:

Country	Mnemonic	Country	Mnemonic
United Kingdom	GBR	Denmark	DNK
Switzerland	CHE	Euro area	EUR
Japan	JPN	Germany	DEU
Canada	CAN	Italy	ITA
Australia	AUS	France	FRA
New Zealand	NZL	Netherlands	NLD
Sweden	SWE	United States	USA
Norway	NOR		

Inflation differential (INF) Difference in inflation rates for a given country and the United

States. Inflation rates are computed from consumer price index (CPI) data from Finaeon

(ticker CP***M); CPI data for the Euro area (EA19) are from the OECD (available at <https://data.oecd.org/price/inflation-cpi.htm>).

Unemployment rate gap differential (UN) Difference in unemployment rate gaps for a given country and the United States. The unemployment rate gap is the cyclical component of the unemployment rate computed using the Christiano and Fitzgerald (2003) band-pass filter for periodicities between six and 96 months. Unemployment rates are from Finaeon (ticker UN***M); the unemployment rate for the Euro area (EA19) is from the OECD (available at <https://data.oecd.org/unemp/unemployment-rate.htm>).

Bill yield differential (BILL) Difference in government bill yields for a given country and the United States. Government bill yields are three-month Treasury bill yields from Finaeon (ticker IT***3D; for the Euro area, IBEUR3D).

Note yield differential (NOTE) Difference in government note yields for a given country and the United States. Government note yields are five-year government bond yields from Finaeon (ticker IG***5D).

Bond yield differential (BOND) Difference in government bond yields for a given country and the United States. Government bond yields are ten-year government bond yields from Finaeon (ticker IG***10D).

Dividend yield differential (DP) Difference in dividend yields for a given country and the United States. Dividend yields are from Finaeon (ticker SY***YM; for the United Kingdom, _DFTASD; for Canada, SYCANYTM; for the Netherlands, SYNLDYAM).

Price-earnings differential (PE) Difference in price-earnings ratios for a given country and the

United States. Price-earnings ratios are from Finaeon (ticker SY***PM; for the United Kingdom, _PFTASD; for Japan, SYJPNPTM; for Canada, SYCANPTM).

Stock market time-series momentum differential (SRET12) Difference in cumulative twelve-month stock market returns for a given country and the United States.

Twelve-month cumulative returns are computed from total return indices from Finaeon.

Finaeon tickers for total return indices are as follows:

Country	Ticker	Country	Ticker
United Kingdom	_TFTASD	Denmark	_OMXCGID
Switzerland	_SSHID	Euro area	_DMIEU0D
Japan	_TOPXDVD	Germany	_CDAXD
Canada	_TRGSPTSE	Italy	_BCIPRD
Australia	_AORDAD	France	_TRSBF250D
New Zealand	_NZGID	Netherlands	_AAXGRD
Sweden	_OMXSBGI	United States	_SPXTRD
Norway	_OSEAXD		

Idiosyncratic volatility (IV) Similarly to Filippou, Gozluklu, and Taylor (2018), we construct month- t IV as follows. We first compute daily log currency excess returns using daily spot and forward rates from Datastream for the 14 countries that we analyze. We then construct daily dollar (MKT_{FX}) and carry trade (HML_{FX}) risk factors for the Lustig et al. (2011) two-factor model. The daily dollar risk factor is the cross-sectional average of the daily log currency excess returns. To construct the daily carry trade risk factor, we first sort the currencies into six portfolios based on the previous day's forward discount and take equally weighted long (short) positions in the currencies in the last (first) portfolio; the carry trade factor is the log return for the long-short portfolio. Each month, we regress daily log

currency excess returns for country i on a constant and the MKT_{FX} and HML_{FX} factors:

$$(1) \quad rx_{t,d}^i = \alpha^i + \beta_{\text{MKT}_{\text{FX},t}}^i \text{MKT}_{\text{FX},t,d} + \beta_{\text{HML}_{\text{FX},t}}^i \text{HML}_{\text{FX},t,d} + \varepsilon_{t,d}^i,$$

where $rx_{t,d}^i$ is the day- d log currency excess return for country i for month t , and $\text{MKT}_{\text{FX},t,d}$ ($\text{HML}_{\text{FX},t,d}$) is the day- d return for month t for the dollar (carry trade) factor. Idiosyncratic volatility is defined as

$$(2) \quad \text{IV}_{i,t} = \left[\frac{1}{T_{i,t}} \sum_{d=1}^{T_{i,t}} (\hat{\varepsilon}_{t,d}^i)^2 \right]^{0.5},$$

where $\hat{\varepsilon}_{t,d}^i$ is the fitted ordinary least squares residual for equation (1), and $T_{i,t}$ is the number of daily log currency excess return observations available for country i for month t .

Idiosyncratic skewness (IS) Defined as

$$(3) \quad \text{IS}_{i,t} = \left(\frac{1}{T_{i,t} - 2} \right) \frac{\sum_{d=1}^{T_{i,t}} (\hat{\varepsilon}_{t,d}^i)^3}{(\text{IV}_{i,t})^3},$$

where $\text{IV}_{i,t}$ is given by equation (2).¹

CPI and unemployment rate data underlying INF and UN are published with a lag. For the United States, both series are published with a one-month lag. Table IA.2 reports publication lags for CPI and unemployment rate data, respectively, for the 14 non-US countries. The last column

¹The construction of idiosyncratic volatility and skewness follows Goyal and Santa-Clara (2003), Fu (2009), Boyer, Mitton, and Vorkink (2010), and Chen and Petkova (2012).

of the table indicates the adjustment made to the timing of the country characteristic to account for the publication lag.

CPI data for Australia and New Zealand are reported at the quarterly frequency and published in the first month of the subsequent quarter. We proceed as follows to compute monthly inflation rates for these two countries that account for the publication lag. Consider, for example, forming a forecast of the exchange rate change for January based on data available through December. Because the CPI is not available for the fourth quarter, we use the CPI inflation for the third quarter divided by three (to convert it to a monthly frequency). When forming the next forecast for February based on data available through January, we use the CPI inflation rate for the fourth quarter divided by three. We use the same inflation rate when forming the forecasts for March and April based on data available through February and March, respectively.

Unemployment rate data for New Zealand are also reported at the quarterly frequency and published in the first month of the subsequent quarter. We proceed in the same manner to compute a monthly unemployment rate for New Zealand that accounts for the publication lag, except that we do not need to divide the quarterly unemployment rate by three.

C. Global Variables

Economic policy uncertainty (EPU) Baker, Bloom, and Davis (2016) Legacy

Three-Component US Economic Policy Uncertainty index based on coverage frequencies in ten major US newspapers (available at https://www.policyuncertainty.com/us_monthly.html).

Monetary policy uncertainty (MPU) Baker et al. (2016) US Monetary Policy Uncertainty

index based on coverage frequencies in ten major US newspapers (available at <https://www.policyuncertainty.com/monetary.html>).

Geopolitical risk (GR) Caldara and Iacoviello (2022) geopolitical risk index based on newspaper coverage (available at <https://www.policyuncertainty.com/gpr.html>).

Global foreign exchange (FX) volatility (GVOL) As in Menkhoff et al. (2012a), month- t global FX volatility is defined as

$$(4) \quad \text{GVOL}_t = \frac{1}{T_t} \sum_{d \in T_t} \left[\sum_{k \in K_{t,d}} \left(\frac{|\Delta s_{t,d}^k|}{K_{t,d}} \right) \right],$$

where $\Delta s_{t,d}^k$ is the day- d log change in the exchange rate for country k and month t , $K_{t,d}$ is the number of currencies available for day d in month t , and T_t is the number of days in month t . We use the 48 countries from Menkhoff et al. (2012a) to compute GVOL.

Global FX illiquidity (GILL) As in Menkhoff et al. (2012a), month- t global FX illiquidity is defined as

$$(5) \quad \text{GILL}_t = \frac{1}{T_t} \sum_{d \in T_t} \left[\sum_{k \in K_{t,d}} \left(\frac{|\text{BAS}_{t,d}^k|}{K_{t,d}} \right) \right],$$

where $\text{BAS}_{t,d}^k$ is the day- d bid-ask exchange rate spread (in percent) for country k and month t . We use the 48 countries from Menkhoff et al. (2012a) to compute GILL.

Global FX correlation (GCOR) Similarly to Mueller, Stathopoulos, and Vedolin (2017),

month- t global FX currency correlation is defined as

$$(6) \quad \text{GCOR}_t = \frac{1}{N_t^{\text{comb}}} \sum_{i=1}^{N_t} \left(\sum_{j>i} \text{RC}_t^{i,j} \right),$$

where $\text{RC}_t^{i,j}$ is the realized correlation between log currency excess returns for countries i and j based on daily data for month t , N_t^{comb} is the number of combinations of currencies (i, j) for month t , and N_t is the number of available currencies for month t . We use the 14 countries in Section IA.1.A to compute GCOR.

IA.2. Deep Neural Network

This section provides details on the deep neural network (DNN) outlined in Section III.B of the paper. A feedforward neural network architecture is comprised of multiple layers. The first is the input layer, which is simply the set of predictors, denoted by $\tilde{x}_1, \dots, \tilde{x}_{P_0}$. One or more hidden layers are next. Each hidden layer l contains P_l neurons, each of which takes predictive signals from the neurons in the previous hidden layer to produce a subsequent signal:

$$(7) \quad h_m^{(l)} = g\left(\omega_{m,0}^{(l)} + \sum_{j=1}^{P_{l-1}} \omega_{m,j}^{(l)} h_j^{(l-1)}\right) \quad \text{for } m = 1, \dots, P_l; l = 1, \dots, L,$$

where $h_m^{(l)}$ is the m th neuron in the l th hidden layer;² $\omega_{m,0}^{(l)}, \dots, \omega_{m,P_{l-1}}^{(l)}$ are weights; and $g(\cdot)$ is a nonlinear activation function. The output layer is a linear function that translates the signals from

²For the first hidden layer, $h_j^{(0)} = \tilde{x}_j$ for $j = 1, \dots, P_0$.

the last hidden layer into a prediction:

$$(8) \quad \hat{y} = \omega_0^{(L+1)} + \sum_{j=1}^{P_L} \omega_j^{(L+1)} h_j^{(L)},$$

where $\hat{y}$ denotes the forecast of the target variable. For the activation function, we use the leaky rectified linear unit (LReLU) function (Maas, Hannun, and Ng (2013)):³

$$(9) \quad g(x) = \begin{cases} x & \text{if } x > 0, \\ 0.01x & \text{if } x \leq 0. \end{cases}$$

Intuitively, equation (9) activates a neuronal connection in response to the strength of the signal, thereby relaying the signal forward through the network.

Training a DNN entails estimating the weights. We estimate the weights by minimizing an objective function based on the training sample MSPE. To better guard against overfitting, we augment the objective function with ℓ_1 and ℓ_2 penalty terms, similarly to equation (5) from the paper. Computationally efficient algorithms based on stochastic gradient descent (SGD) are available to estimate the DNN weights. We use the powerful Adam algorithm (Kingma and Ba (2015)). Due to the stochastic nature of the SGD algorithm, the estimated weights depend on the seed for the random number generator. To reduce the dependency of the DNN forecast on the seed, we train the model three times with different seeds and take an average of the forecasts across the three fitted DNNs.

Given our noisy data environment, we take additional steps to further guard against

³The LReLU refines the conventional ReLU (which sets $g(x) = 0$ if $x \leq 0$) to help prevent the DNN from “dying” during training, meaning that the neurons in the DNN never activate.

overfitting the DNN. We set the intercept terms for the weights ($\omega_{m,0}^{(l)}$ for $m = 1, \dots, P_l$; $l = 1, \dots, L$ and $\omega_0^{(L+1)}$) to zero.⁴ These restrictions are analogous to the absence of an intercept term in the baseline panel predictive regression in equation (1) from the paper, ensuring that the DNN predicts a value of zero for the exchange rate change when all of the (standardized) predictors are at their mean value of zero. We continue to pool the data in the DNN to impose the homogeneity restrictions that the weights are the same across countries. Imposing both the zero intercept and slope homogeneity restrictions substantially reduces the number of weights that we need to estimate, thereby helping to guard against overfitting. In addition, we employ dropout (Srivastava, Hinton, Krizhevsky, Sutskever, and Salakhutdinov (2014)), which randomly drops a portion of the neurons in a hidden layer when training the model via the SGD algorithm and is helpful in mitigating overfitting.

We tune several hyperparameters for the DNN—including the ℓ_1 and ℓ_2 regularization hyperparameters, dropout rate, and batch size—by considering a grid of values for each and using observations for the last 30% of months for the available data at the time of forecast formation as a validation sample, thereby respecting the time-series dimension of the panel data. We select the vector of hyperparameter values that minimizes the objective function for the validation sample. To help keep the hyperparameters relevant, we tune them every 36 months.

⁴The intercept terms for the weights are typically called “bias” terms in the machine-learning literature, although it does not have its traditional econometric meaning in this context.

IA.3. Non-US Dollar Base Currencies

As shown in Section IV.C of the paper, our approach performs well when the US dollar is used as the base currency, which is consistent with its dominant role as the primary settlement currency in international trade and global financial markets. In this section, we consider alternative base currencies, namely, the Japanese yen and Swiss franc. For the former (latter), the exchange rate is the number of country- i currency units per Japanese yen (Swiss franc). We measure the country characteristics as the country- i value minus the Japanese or Swiss value. For each of the alternative base currencies, we generate out-of-sample forecasts for 1995:01 to 2020:09 as described in the paper.

Tables IA.5 and IA.6 report results when the Japanese yen and Swiss franc, respectively, serve as the base currency. Our machine-learning approach also works reasonably well for forecasting exchange rate changes in Table IA.5 when the Japanese yen serves as the base currency. However, compared to the results reported in Table 1 from the paper, the degree of exchange rate predictability is not as strong as when the US dollar is the base currency. When the Swiss franc serves as the base currency, the exchange rate forecasts are substantially less accurate in Table IA.6 compared to Table 1 from the paper. Overall, we find less predictive power when other currencies, such as the Japanese yen and Swiss franc, serve as the base currency. As mentioned in Section IV.C of the paper, understanding the economic reasons for such differences is an interesting avenue for future research.

IA.4. Robustness Checks: Forecast Accuracy

A. Elastic Net Hyperparameter Tuning

Conventional M -fold cross validation (e.g., as implemented in the `glmnet` package in R) randomly divides the observations for the training sample into M non-overlapping and generally non-contiguous blocks or folds of approximately equal size. For each combination of δ and λ in a two-way grid, we first fit the baseline predictive regression in equation (1) from the paper via the elastic net (ENet, Zou and Hastie (2005)) using the available observations after excluding those in the first fold; for each combination of δ and λ , we use the fitted model to compute predictions for the observations in the first fold and calculate the mean squared prediction error (MSPE). Next, for each combination of δ and λ , we fit the model via the ENet using available observations after excluding those in the second fold; for each combination of δ and λ , we use the fitted model to compute predictions for the observations in the second fold and calculate the MSPE. We continue in this fashion through the remaining folds. Finally, for each combination of δ and λ , we compute the average of the MSPEs across the M folds and select the combination of δ and λ that minimizes the average MSPE.

A potential drawback to conventional M -fold cross validation in our context is that it does not take into account the time-series nature of panel data. Specifically, models can be fitted using observations that temporally follow those in a validation fold. To address this issue, we consider additional tuning methods based on validation in a panel data context. We refer to the first as time-series validation. Suppose that we want to generate forecasts for month $t + 1$ using data available through month t . We divide the panel data into observations corresponding to the first 70% and last 30% of months from the start of the sample through month t , where observations

from the last 30% of months comprise the validation period. The split preserves the temporal ordering of the data. For each combination of δ and λ , we fit the baseline predictive regression in equation (1) from the paper via the ENet using data for the first 70% of months, use the fitted model to compute predictions for the validation period, and calculate the MSPE. We select the combination of δ and λ that minimizes the MSPE.

To implement cross validation (i.e., multiple validation periods) in a manner that recognizes the time-series nature of panel data, we implement what we call time-series cross validation in a panel data context. Again suppose that we want to generate forecasts for month $t + 1$ using data available through month t . We begin by using panel data observations for the first 25% of months from the start of the sample through month t to fit the baseline predictive regression in equation (1) from the paper via the ENet for each combination of δ and λ . For each combination of δ and λ , we then use the fitted model to compute predictions for panel data observations for the next 25% of months (first fold) and calculate the MSPE. Next, we use panel data for the first 50% of months from the start of the sample through month t to fit the model via the ENet for each combination of δ and λ ; for each combination of δ and λ , we use the fitted model to compute predictions for panel data observations for the next 25% of months (second fold) and calculate the MSPE. In the last iteration, we use panel data for the first 75% of months from the start of the sample through month t to fit the model via the ENet for each combination of δ and λ ; for each combination of δ and λ , we use the fitted model to compute predictions for panel data observations for the last 25% of months (third fold) and calculate the MSPE. Finally, for each combination of δ and λ , we compute the average of the MSPEs across the three folds and select the combination of δ and λ that minimizes the average MSPE.

Table IA.7 reports out-of-sample R^2 statistics for baseline ENet forecasts that use

different methods for tuning δ and λ . For all methods, we retune the hyperparameters every month. The values in the fourth through seventh columns are typically less than the corresponding values in the fifth column of Table 1 from the paper.

B. Random Features Forecast

We also consider forecasts based on the random features model in Kelly, Malamud, and Zhou (2024), which is essentially a neural network with one hidden layer (with a very large number of nodes) and random weights for the hidden layer. As shown in Table IA.8, the random features forecasts do not perform as well as the DNN forecast in the seventh column of Table 1 from the paper.

C. Ridge Forecast

Table IA.9 reports out-of-sample R^2 statistics for a forecast based on ridge estimation of the baseline predictive regression in equation (1) from the paper as well as an ensemble forecast comprised of the baseline ridge and DNN forecasts. Overall, the ridge-based forecasts are less accurate than the corresponding baseline ENet and ensemble forecasts reported in Table 1 from the paper, so allowing for sparsity provides better out-of-sample performance for our application.

D. Country-Specific Data

We also generate baseline ENet, DNN, and ensemble forecasts by estimating models using country-specific data. As shown in Table IA.10, forecasts generated using country-specific data are nearly always substantially less accurate than the forecasts based on pooling in Table 1

from the paper. Thus, for out-of-sample time-series exchange rate prediction, information appears better processed via homogeneity of the model parameters across countries in light of the bias-variance trade-off.

E. Random Forests

In addition to the baseline ENet and DNN forecasts, we consider a popular machine-learning method based on decision trees, random forests (Breiman (2001)). Like DNNs, random forests are powerful machine-learning techniques that perform well in a variety of domains. However, although they perform reasonably well, random forests are not as effective for forecasting monthly exchange rates as ENet estimation of the baseline predictive regression and the DNN. We explain why this is the case.

A random forest is built up from regression trees, which are machine-learning devices that incorporate nonlinearities via multi-way interactions and higher-order effects of the predictors. To construct a regression tree, the predictor space is sequentially split into regions, typically using the classification and regression tree (CART) algorithm (Breiman, Friedman, Stone, and Olshen (1984)). The final set of regions is referred to as “terminal nodes” or “leaves,” and the prediction is the average value of the target in a given leaf. The forecast corresponding to a regression tree with U leaves can be expressed as

$$(10) \quad \widehat{\Delta s_{i,T+1|T}}^{\text{RT}} = \sum_{u=1}^U \overline{\Delta s_u} \mathbf{1}_u(\tilde{\mathbf{x}}_{i,t}; \boldsymbol{\eta}_u),$$

where the indicator function $\mathbf{1}_u(\tilde{\mathbf{x}}_{i,t}; \boldsymbol{\eta}_u) = 1$ if $\tilde{\mathbf{x}}_{i,t} \in R_u(\boldsymbol{\eta}_u)$ for the u th region denoted by R_u

(which is determined by the parameter vector η_u) and zero otherwise, and $\overline{\Delta s_u}$ is the average value of the target observations in R_u for the training sample based on data through T .

A large (i.e., “deep”) regression tree can often capture complex nonlinear relations in the data, but it is susceptible to overfitting due to the high variance of the tree in light of the bias-variance trade-off. To reduce the variance, a random forest averages forecasts across many deep regression trees, where each tree is constructed based on a bootstrap sample of the original data using a randomly selected subset of the predictors for each split. Using a randomly selected subset of the predictors “decorrelates” the trees to further reduce the variance. Indexing the bootstrap samples by b , the random forest forecast can be expressed as

$$(11) \quad \widehat{\Delta s_{i,T+1:T}}^{\text{RF}} = \frac{1}{B} \sum_{b=1}^B \left[\sum_{u=1}^U \overline{\Delta s_u}^{(b)} \mathbf{1}_u^{(b)}(\tilde{\mathbf{x}}_{i,t}; \eta_u) \right],$$

where B is the number of bootstrap samples, and $\overline{\Delta s_u}^{(b)}$ and $\mathbf{1}_u^{(b)}(\tilde{\mathbf{x}}_{i,t}; \eta_u)$ are the analogs to $\overline{\Delta s_u}$ and $\mathbf{1}_u(\tilde{\mathbf{x}}_{i,t}; \eta_u)$, respectively, in equation (10) for the b th bootstrap sample. As with the DNN, we tune various hyperparameters for the random forest by considering a grid of values for each and using observations for the last 30% of months for the available data at the time of forecast formation as a validation sample and select the vector of hyperparameter values that minimizes the objective function for the validation sample.⁵

Table IA.11 reports out-of-sample R^2 statistics for the random forest forecast as well as an ensemble-all forecast that is an average of the baseline ENet, DNN, and random forest forecasts.

⁵We fit the random forests using the ranger package in R. We tune the following set of hyperparameters for the random forest as defined in the ranger package documentation: “num.trees,” “mtry,” “min.node.size,” and “sample.fraction.”

Overall, the random forest forecast performs reasonably well. The $R^2_{i,OS}$ statistics are positive for eight of the ten individual countries, and all positive $R^2_{i,OS}$ statistics are significant at conventional levels. Taking all of the countries together, the $R^2_{All,OS}$ statistic is 1.10% for the random forest, which is significant at the 1% level.

Compared to the baseline ENet and DNN forecasts in Table 1 from the paper, the random forest forecast in Table IA.11 generally does not perform as well, as the $R^2_{All,OS}$ statistics for the baseline ENet and DNN forecasts are larger vis-à-vis that for the random forest. A likely explanation for the better overall relative performance of the baseline ENet and DNN forecasts is due to our imposition of the zero restrictions for the country mean exchange rate changes for these forecasts, as described in Section III of the paper. The restrictions are in line with the data and motivated by the stringency of the no-change benchmark forecast in the literature. Given the noise in monthly exchange rate changes, restricting the country means to zero is highly beneficial in helping to prevent overfitting in light of the bias-variance trade-off. However, because of the nature of decision trees, restricting the country mean exchange rates to zero is not possible, so we are unable to refine the random forest forecast in this manner, which generally adversely affects out-of-sample performance compared to the baseline ENet and DNN forecasts.⁶

⁶It is interesting to note that the ensemble-all forecast in the last column of Table IA.11, which is a combination of the baseline ENet, DNN, and random forest forecasts, produces an $R^2_{All,OS}$ statistic of 2.13% (significant at the 1% level) in the last row of Table IA.11, which is very close to the corresponding $R^2_{All,OS}$ statistic (1.86%) for the ensemble forecast based on the baseline ENet and DNN forecasts in Table 1 from the paper.

F. Additional Predictors

There are additional predictors of exchange rate changes proposed in the literature that we do not include in our baseline analysis. For example, Della Corte, Jeanneret, and Patelli (2023) consider the credit-implied risk premium for 16 Euro-area countries, showing that it predicts US dollar-euro exchange rate changes on an in-sample basis (but they do not consider out-of-sample forecast accuracy). However, this measure is confined to the Euro area, so it does not provide sufficient coverage for our sample.

Various studies document a link between commodity returns and exchange rates. A number of these studies find that exchange rate changes predict commodity returns (e.g., Chen, Rogoff, and Rossi (2010)), while fewer find that commodity returns contain information about future exchange rate changes (e.g., Kohlscheen, Avalos, and Schrimpf (2017)). To account for the latter, we augment our baseline specification with a measure of commodity returns and assess its incremental predictive power.

To this end, we begin by constructing a country-level commodity price index following the methodology of Gruss and Kebhaj (2019). For each country and month, we compute a weighted average of log commodity price changes using each commodity's net export share as the weight, with the weights lagged two years to account for publication delays. This yields a terms-of-trade index that we label CXP, which can be viewed as a measure of net commodity returns. For the Euro area, we aggregate the indices of Germany, France, Italy, and the Netherlands using time-varying GDP weights, treating the bloc as a single pseudo-country from January 1999 onward. We compute the difference between the CXP for country i and the United

States. CXP enters our panel of data as an additional country characteristic for forecasting the log exchange rate change.

As a preliminary analysis, we examine (in-sample) dynamic links between exchange rate changes and CXP using Granger causality tests. Panel A of Table IA.12 reports results using country-level data with the lag orders selected via the AIC (considering a maximum of twelve), while Panel B provides results for various assumed lag orders; finally, Panel C reports results for the Dumitrescu and Hurlin (2012) panel test of Granger causality. Overall, the evidence in Table IA.12 strongly favors a unidirectional relation from exchange rate changes to net commodity returns, consistent with the prevailing view in the literature for developed countries. In Panel A, the null hypotheses that the exchange rate change does not Granger cause CXP is rejected for nine of 14 countries at conventional significance levels; the reverse null, that CXP does not Granger cause the exchange rate change, is only rejected for two countries (Canada and France at the 5% and 10% levels, respectively). Panel B confirms the asymmetry across lag orders of one, three, six, and twelve months: between 43% and 57% of countries exhibit significant Granger causality from the exchange rate change to CXP, compared to at most 14% in the reverse direction. Panel C further reinforces these findings at the panel level, with the Dumitrescu and Hurlin (2012) $\bar{Z}$ statistics rejecting the null that the exchange rate change does not Granger cause CXP at the 1% level for all reported lag orders, while none of the statistics are significant for the reverse null.

Given that the results in Table IA.12 point to a dominant flow of information from exchange rate changes to net commodity returns (rather than the reverse), the incremental forecasting ability of CXP for exchange rate changes may be limited. To investigate this issue, Table IA.13 reports out-of-sample R^2 statistics when CXP is included among the set of country characteristics. Consistent with the Granger causality evidence, the out-of-sample gains relative to

the results in Table 1 from the paper are modest and uneven across countries. The baseline ENet, DNN, and ensemble forecasts provide positive and often statistically significant improvements in forecast accuracy for commodity-exporting economies such as Australia, New Zealand, Sweden, and Norway. For a non-commodity-exporting economy such as Japan, the out-of-sample R^2 statistics are all negative in the last three columns of Table IA.13. The composite out-of-sample R^2 statistics in the last row produce statistically significant improvements in forecast accuracy of 1.07%, 0.92%, and 1.27% for the baseline ENet, DNN, and ensemble forecasts, respectively. However, these statistics do not exceed those in Table 1 from the paper, in line with the literature indicating that exchange rate changes are more important for predicting net commodity returns. Thus, we omit CXP from the main analysis in the paper.⁷

G. Alternative Benchmarks

We check the robustness of the results in Table 1 from the paper against two other benchmarks based on popular predictors from the literature. The first benchmark uses Taylor (1993) rule fundamentals. Specifically, we compute a forecast based on a linear panel predictive regression that uses the inflation differential and unemployment rate gap differential as predictors and is fitted via OLS. The second is a forecast computed from a linear panel predictive regression that uses the inflation differential and bill yield differential as predictors and again is fitted via OLS. The latter is motivated by recent studies that include similar variables in predictive regressions when analyzing currency markets (e.g., Chernov and Creal (2021), Eichenbaum, Johannsen, and Rebelo (2021), Dahlquist and Pénasse (2022), Chernov, Dahlquist, and Lochstoer

⁷Our results are in line with Jeanneret and Sokolovski (2026), who find that exchange rate predictability based on changes in country-specific commodity export or import prices is limited to illiquid emerging currencies.

(2023)).⁸ As shown in Tables IA.14 and IA.15, the results are very similar to those in Table 1 from the paper, as the baseline ENet, DNN, and ensemble forecasts significantly outperform the alternative benchmarks for nearly all individual countries, often by a substantial margin, as well as when the ten countries are considered jointly.

H. Cumulative Differences in Squared Prediction Errors

To provide additional perspective on the consistency of the out-of-sample gains in Figure 1 from the paper, Figure IA.6 employs the graphical device of Goyal and Welch (2008). The figure portrays cumulative differences in squared prediction errors for the no-change benchmark forecast vis-à-vis the baseline ENet, DNN, and ensemble forecasts (considered in turn). Each curve conveniently allows for a comparison of forecast accuracy in terms of MSPE for any subsample: if the curve is higher (lower) at the end of the segment corresponding to the subsample, then the competing (benchmark) forecast has a lower MSPE for the subsample. Thus, a forecast that always outperforms the benchmark will have a curve with a uniformly positive slope. Of course, given that exchange rate changes have a large unpredictable component, this ideal is practically unattainable. Realistically, we seek a forecast with a curve that is predominantly positively sloped and does not have extended segments with negative slopes.

According to Figure IA.6, the curves are positively sloped for much of the time for the different countries, so the baseline ENet, DNN, and ensemble forecasts outperform the no-change benchmark on a consistent basis over time. By blending the baseline ENet and DNN forecasts, the ensemble approach generally provides the most consistent out-of-sample gains. The

⁸We obtain similar results when we use the real exchange rate in place of the inflation differential.

improvements in accuracy provided by the baseline ENet, DNN, and ensemble forecasts vis-à-vis the no-change benchmark are often particularly strong during the worst phase of the Global Financial Crisis in late 2008, so the information in the predictors becomes especially important during the crisis. Sizable gains are also evident before the crisis in a number of cases, and the gains are quite consistent and substantive for many countries after 2008 through the end of the out-of-sample period. Compared to the baseline ENet forecast, the DNN forecast substantially improves performance around the crisis for Switzerland, Japan, and Canada.

I. Safe-Haven Currency

A spate of recent papers (e.g., Kremens and Martin (2019), Adrian and Xie (2020), Jiang, Krishnamurthy, and Lustig (2021), Lilley, Maggiori, Neiman, and Schreger (2022)) finds evidence of exchange rate predictability around the Global Financial Crisis based on the US dollar's perception as a safe-haven currency. Due to data availability, these studies analyze predictability at a quarterly horizon or longer and/or employ relatively short samples. Although we focus on a monthly horizon and consider a longer out-of-sample period, our machine-learning forecasts appear to capture exchange rate predictability around the crisis relating to a safe-haven role for the US dollar. Specifically, the baseline ENet, DNN, and ensemble forecasts in Figures IA.2, IA.3, and IA.4, respectively, portend strong depreciations for many countries' currencies in late 2008 during the worst phase of the crisis; as shown in Figure 1 from the paper and Figure IA.6, the machine-learning forecasts substantially outperform the no-change benchmark during that time.

We also examine links between the set of predictors selected by the baseline ENet and the Lilley et al. (2022) capital flow measure, which is available at the quarterly frequency and is

based on the change in US holdings of foreign bonds. Lilley et al. (2022) find that their measure is significantly related to various proxies for global risk appetite, as well as the change in a broad US dollar index from 2007 to 2017, which they interpret as evidence of the US dollar's role as a safe-haven currency during the crisis. We investigate links between the ten predictors selected by the ENet in the fitted baseline panel predictive regression based on data through 2020:08 (corresponding to the forecast for the final month, 2020:09) and the change in US foreign bond holdings.⁹ We average the ten predictors over the three months comprising a quarter, as well as across countries. We then regress the change in US foreign bond holdings on the set of ten time- and country-aggregated predictors for 2007:1 to 2019:2.¹⁰

Figure IA.7 shows the (standardized) change in US foreign bond holdings, together with the fitted values for the regression. As a group, the ten predictors selected by the ENet in the fitted baseline panel predictive regression are significantly related to the change in US foreign bond holdings at the 5% level, and the predictors collectively explain 38% of the variation in the capital flow measure. The fitted values track the actual values quite closely during the Global Financial Crisis, providing further evidence that relevant predictors selected by the ENet contain information pertaining to a safe-haven role for the US dollar.

As discussed in Section IA.3, we also computed exchange rate forecasts with the Japanese yen replacing the US dollar as the base currency. While our machine-learning approach also works reasonably well for forecasting exchange rate changes with the Japanese yen as the base

⁹The ten predictors selected by the ENet are UN, DP, PE, INF.GVOL, UN.GVOL, BILL.GVOL, PE.EPU, PE.GILL, SRET12.GR, and IV.GCOR.

¹⁰Data for the change in US foreign bond holdings from Lilley et al. (2022) are available from the Global Capital Allocation Project website.

currency (see Table IA.5), the degree of exchange rate predictability is not as strong as when the US dollar is the base currency in Table 1 from the paper. This provides further evidence of a safe-haven (and/or dominant currency) role for the US dollar.

IA.5. Model Interpretation

A. Recursive Baseline ENet Estimates

As a complement to Figure 2 from the paper, Figure IA.9 depicts recursive ENet estimates of the slope coefficients for the fitted baseline panel predictive regressions used to generate the out-of-sample baseline ENet forecasts for the 1995:01 to 2020:09 out-of-sample period. For the full sample, recall that the ENet selects UN, DP, PE, INF.GVOL, UN.GVOL, BILL.GVOL, PE.EPU, PE.GILL, SRET12.GR, and IV.GCOR. Among these predictors, Figure IA.9 shows that DP, BILL.GVOL, and SRET12.GR are selected the vast majority of the time over the out-of-sample period, and the signs of their coefficient estimates always match those for the full-sample estimates. The magnitude of coefficient estimate for DP is fairly consistent over time, while that for BILL.GVOL displays a general upward trend. The coefficient for SRET12.GR increases in magnitude in the early 2000s and remains reasonably sizable through the rest of the out-of-sample period.

Turning to the other predictors selected by the ENet for the full sample, UN is consistently selected from around the mid 2000s through 2010. Thereafter, it is selected less often until toward the end of the out-of-sample period. The magnitude of the coefficient estimate for UN is largest near the Great Recession as well as the recession in early 2020 corresponding to the COVID-19

crisis. PE is selected on a regular basis starting in the early 2000s, and its coefficient estimate remains limited in magnitude over time. INF.GVOL is selected starting near the end of Great Recession and continues to be selected thereafter, with the coefficient estimate evincing a slight upward trend. While UN.GVOL is always selected from the early 2000s through the early 2010s, and its coefficient estimate is typically sizable in magnitude (especially in the wake of the Great Recession), it is only infrequently selected outside of this period, and the magnitude of the estimate is considerably smaller. PE.EPU and PE.GILL are selected starting in 2009, with the former (latter) selected sporadically (regularly) through the end of the out-of-sample period, and the magnitudes of their coefficient estimates are typically limited. Beginning in the early 2000s, IV.GCOR is almost always selected, and its coefficient estimate is quite sizable from the early 2000s through 2010 and again near the end of the out-of-sample period.

There are some predictors that are not selected by the ENet for the full sample in Figure 2 from the paper although they are earlier in the out-of-sample period in Figure IA.9. These include IV, NOTE.GVOL, and DP.GCOR, which are selected fairly frequently, as well as INF.GCOR, UN.EPU, BILL.GCOR, DP.GVOL, SRET12.MPU, IV.GR, and IV.GVOL, which are selected very rarely (and their coefficient estimates are very small in magnitude). The remaining 50 predictors are never selected by the ENet over the out-of-sample period. Figure IA.9 helps to explain the benefits of the ENet for improving out-of-sample exchange rate prediction. We cannot know a priori which country characteristics (or fundamentals) and their interactions with global variables are the most relevant for predicting exchange rate changes. Thus, we consider a large set of country characteristics and interactions, resulting in a large-dimensional predictive regression. However, estimating a large dimensional model, in conjunction with an inherently small signal-to-noise ratio, renders conventional OLS estimation highly susceptible to in-sample

overfitting and thus poor out-of-sample performance. To mitigate overfitting, we employ regularized regression via the ENet, which shrinks the coefficient estimates, many of them to exactly zero, to produce a sparse model that focuses on the most relevant predictors. As shown in Section IV.C of the paper, the strategy of considering a large number of country characteristics in conjunction with machine-learning methods produces significant improvements in forecast accuracy when it comes to out-of-sample prediction of short-run exchange rate changes. According to Figure IA.9, together, UN, DP, PE, BILL.GVOL, SRET12.GR, and IV.GCOR are all consistently selected by the ENet since the early 2000s; among these predictors, DP, BILL.GVOL, SRET12.GR, and IV.GCOR also exhibit relatively large coefficient estimates on a regular basis, identifying them as the most relevant predictors of short-run exchange rate changes over the out-of-sample period.

B. DNN and Ensemble Forecasts

The ENet is a machine-learning device that facilitates model interpretation by performing model selection, and the ENet coefficient estimates for the baseline panel predictive regression in Figure 2 from the paper and Figure IA.9 are tantamount to variable-importance measures. However, when we model the conditional expectation function in equation (6) from the paper using the DNN in Section III.B of the paper, the fitted model becomes a “black box.” Thus, we need to employ model-interpretation tools developed in the machine-learning literature. For the fitted DNNs used to generate the out-of-sample DNN and ensemble forecasts, we measure variable importance using the approach of Greenwell, Boehmke, and McCarthy (2018), which is based on partial dependence plots (PDPs, Friedman (2001)).

A PDP measures the relation between the expected value of the dependent variable and a given predictor for any fitted model, including “black-box” models like DNNs. Let $\hat{f}(\mathbf{x})$ denote the prediction function for a generic fitted model, where $\mathbf{x}$ denotes the vector of predictors. The PDP for predictor x_q is defined as

$$(12) \quad \begin{aligned} f_q(x_q) &= E_{\mathbf{x}_{C(q)}} [\hat{f}(x_q, \mathbf{x}_{C(q)})] \\ &= \int_{\mathbf{x}_{C(q)}} \hat{f}(x_q, \mathbf{x}_{C(q)}) p_{C(q)}(\mathbf{x}_{C(q)}) d\mathbf{x}_{C(q)}, \end{aligned}$$

where $\mathbf{x}_{C(q)} = \mathbf{x} \setminus x_q$, and

$$(13) \quad p_{C(q)}(\mathbf{x}_{C(q)}) = \int_{x_q} p(x_q, \mathbf{x}_{C(q)}) dx_q$$

is the joint marginal probability density for $\mathbf{x}_{C(q)}$. Equation (12) is typically estimated via Monte Carlo integration using the training data, which in our panel data setting is given by

$$(14) \quad \bar{f}_q(x_q) = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \hat{f}(x_q, \mathbf{x}_{i,t,C(q)}).$$

It is common to analyze the PDP for a grid of x_q values in the range of training-sample observations, which we denote by $\{x_{q,j}\}_{j=1}^J$. For the ensemble of models, the PDP is the average of the measures in equation (14) across the fitted models comprising the ensemble.

Greenwell et al. (2018) propose a PDP-based measure of variable (or predictor)

importance:

$$(15) \quad \mathcal{I}(x_q) = \left\{ \left[1/(J-1) \right] \sum_{j=1}^J [\bar{f}_q(x_{q,j}) - (1/J) \sum_{j=1}^J \bar{f}_q(x_{q,j})]^2 \right\}^{0.5}.$$

Intuitively, equation (15) uses the sample standard deviation of the PDP to measure its “flatness.”

If the expected value of the target does not change as the predictor changes, then the PDP is flat, so the variable importance measure is zero. As the variability of the PDP increases, the importance measure in equation (15) likewise increases.

Figures IA.10 and IA.11 depict PDP-based variable-importance measures for the sequence of fitted prediction models underlying the DNN and ensemble forecasts, respectively. In terms of variable importance, the figures paint a broadly similar picture to that for the recursive baseline ENet estimates in Figure IA.9, including the crucial role of a number of interaction terms.¹¹

IA.6. A Closer Look at Regime Switching

Section III.A of the paper explains how the interaction terms in the baseline panel predictive regression in equation (1) from the paper allow for continuous regime switching in the response of the exchange rate change to a change in a country characteristic; see equation (2) from the paper. Section V of the paper shows that the interaction terms, especially BILL.GVOL, play important roles in the fitted prediction models. Motivated by our findings, as well as evidence in Clarida, Davis, and Pedersen (2009), we explore regime switching in more detail in

¹¹The variable-importance measures in Figures IA.10 and IA.11 are always positive by construction. They can be compared to the absolute values of the recursive baseline ENet estimates in Figure IA.9.

this section, focusing on GVOL as the global variable governing regime switching. We augment the interaction terms with threshold or logistic smooth transition mechanisms, thereby considering richer forms of regime switching. For consistency, we continue to specify the panel predictive regressions in terms of standardized predictors.

A. Interaction

Consider the following panel predictive regression with an interaction term:

$$(16) \quad \Delta s_{i,t+1} = \underbrace{\beta^z \frac{1}{\sigma_i^z} (z_{i,t} - \mu_i^z)}_{\tilde{z}_{i,t}} + \underbrace{\beta^{z,g} \frac{1}{\sigma_i^{z,g}} (z_{i,t} g_t - \mu_i^{z,g})}_{\tilde{z}_{i,t} g_t} + \varepsilon_{i,t+1},$$

where μ_i^z ($\mu_i^{z,g}$) is the mean of $z_{i,t}$ ($z_{i,t} g_t$), and σ_i^z ($\sigma_i^{z,g}$) is the standard deviation of $z_{i,t}$ ($z_{i,t} g_t$).

For notational simplicity, we consider only a single country characteristic and a single global variable; it is straightforward to extend equation (16) to include additional country characteristics and global variables, as in equation (1) from the paper. We can re-write equation (16) as

$$(17) \quad \Delta s_{i,t+1} = \underbrace{\left(\beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} g_t \right)}_{\text{slope coefficient for } z_{i,t}} z_{i,t} - \left(\beta^z \frac{1}{\sigma_i^z} \mu_i^z + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} \mu_i^{z,g} \right) + \varepsilon_{i,t+1}.$$

Equation (17) shows that the inclusion of the interaction term in equation (16) is equivalent to allowing for the slope coefficient for $z_{i,t}$ to vary continuously with g_t in a linear manner, which is what we term continuous regime switching in Section III.A of the paper. To compare different regime-switching specifications, as in equation (2) from the paper, it is useful to consider $\frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}}$,

which, for equation (17), is given by

$$(18) \quad \frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}} = \beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} g_t.$$

If $\beta^{z,g} > 0$, then $\frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}}$ increases linearly with g_t . For example, if $z_{i,t}$ is $\text{BILL}_{i,t}$, g_t is GVOL_t , and $\beta^{z,g} > 0$, then the slope coefficient for $\text{BILL}_{i,t}$ increases continuously with GVOL_t , as discussed in Section V of the paper.

B. Interaction with Threshold

Next, we consider an interaction specification with a threshold mechanism:

$$(19) \quad \begin{aligned} \Delta s_{i,t+1} &= \underbrace{\beta^z \frac{1}{\sigma_i^z} (z_{i,t} - \mu_i^z)}_{\tilde{z}_{i,t}} + \underbrace{\beta^{z,g} \frac{1}{\sigma_i^{z,g}} [z_{i,t} g_t \mathbb{I}(g_t; g^*) - \mu_i^{z,g}]}_{z_{i,t} g_t \mathbb{I}(g_t; g^*)} + \varepsilon_{i,t+1} \\ &= \underbrace{\left[\beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} g_t \mathbb{I}(g_t; g^*) \right]}_{\text{slope coefficient for } z_{i,t}} z_{i,t} - \left(\beta^z \frac{1}{\sigma_i^z} \mu_i^z + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} \mu_i^{z,g} \right) + \varepsilon_{i,t+1}, \end{aligned}$$

where $\mathbb{I}(g_t; g^*) = 1$ if $g_t \geq g^*$ and zero otherwise, g^* is the threshold parameter for the global variable, and $\mu_i^{z,g}$ ($\sigma_i^{z,g}$) is the mean (standard deviation) of $z_{i,t} g_t \mathbb{I}(g_t; g^*)$. Based on equation (19),

$$(20) \quad \frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}} = \beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} g_t \mathbb{I}(g_t; g^*).$$

According to equation (20) and assuming that $\beta^{z,g} > 0$, $\frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}}$ is constant at $\beta^z \frac{1}{\sigma_i^z}$ for $g_t < g^*$, jumps to $\beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} g^*$ at $g_t = g^*$, and increases linearly with g_t (with a slope of $\beta^{z,g} \frac{1}{\sigma_i^{z,g}}$) for

$g_t > g^*$. If we again assume that $z_{i,t}$ is $\text{BILL}_{i,t}$, g_t is GVOL_t , and $\beta^{z,g} > 0$, then the slope coefficient for $\text{BILL}_{i,t}$ is constant for $\text{GVOL}_t < \text{GVOL}^*$, jumps to a higher value when $\text{GVOL}_t = \text{GVOL}^*$, and increases continuously with GVOL_t for $\text{GVOL}_t > \text{GVOL}^*$.¹²

C. Interaction with Logistic Smooth Transition

Analogously to combining the interaction specification with a threshold mechanism in equation (19), we also consider combining the interaction specification with a logistic smooth transition (LST) mechanism:

$$\begin{aligned}
 \Delta s_{i,t+1} &= \underbrace{\beta^z \frac{1}{\sigma_i^z} (z_{i,t} - \mu_i^z)}_{\tilde{z}_{i,t}} + \underbrace{\beta^{z,g} \frac{1}{\sigma_i^{z,g}} [z_{i,t} g_t \mathbb{L}(g_t; \kappa) - \mu_i^{z,g}]}_{\underbrace{z_{i,t} g_t \mathbb{L}(g_t; \kappa)}} + \varepsilon_{i,t+1} \\
 (21) \quad &= \underbrace{\left[\beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} g_t \mathbb{L}(g_t; \kappa) \right]}_{\text{slope coefficient for } z_{i,t}} z_{i,t} - \left(\beta^z \frac{1}{\sigma_i^z} \mu_i^z + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} \mu_i^{z,g} \right) + \varepsilon_{i,t+1},
 \end{aligned}$$

where $\mathbb{L}(g_t; \kappa) = [1 + \exp(-\kappa \frac{1}{\sigma_g}(g_t - \mu_g))]^{-1}$ for $\kappa > 0$ is the logistic function, μ_g (σ_g) is the mean (standard deviation) of g_t , and $\mu_i^{z,g}$ ($\sigma_i^{z,g}$) is the mean (standard deviation) of $z_{i,t} g_t \mathbb{L}(g_t; \kappa)$.

The curvature parameter κ controls how sharply the transition occurs around μ_g : larger values of κ produce a sharper transition, while smaller values produce a more gradual one. Based on

¹²When $z_{i,t} g_t \mathbb{L}(g_t; g^*)$ is simplified to $z_{i,t} \mathbb{L}(g_t; g^*)$ in equation (19), the panel predictive regression becomes a conventional threshold specification. In this case, $\frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}}$ is a step function: the slope coefficient equals $\beta^z \frac{1}{\sigma_i^z}$ ($\beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}}$) for $g_t < g^*$ ($g_t \geq g^*$).

equation (21),

$$(22) \quad \frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}} = \beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}} g_t \mathbb{L}(g_t; \kappa).$$

As $g_t \rightarrow -\infty$, $\mathbb{L}(g_t; \kappa) \rightarrow 0$, so $\frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}}$ approaches $\beta^z \frac{1}{\sigma_i^z}$. For $\beta^{z,g} > 0$, as g_t increases, $\frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}}$ in equation (22) smoothly increases according to $g_t \mathbb{L}(g_t; \kappa)$. As $g_t \rightarrow \infty$, $\mathbb{L}(g_t; \kappa) \rightarrow 1$; thus, in the limit, $\frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}}$ increases linearly as g_t increases (with slope $\beta^{z,g} \frac{1}{\sigma_i^{z,g}}$). In addition, as $\kappa \rightarrow \infty$, $\frac{\partial \Delta s_{i,t+1}}{\partial z_{i,t}}$ in equation (22) approaches the expression in equation (20) (for $g^* = \mu_g$). Continuing to assume that $z_{i,t}$ is $\text{BILL}_{i,t}$, g_t is GVOL_t , and $\beta^{z,g} > 0$, the slope coefficient for $\text{BILL}_{i,t}$ approaches a constant value as $\text{GVOL}_t \rightarrow -\infty$; as GVOL_t increases, the slope coefficient continuously increases according to $\text{GVOL}_t \mathbb{L}(\text{GVOL}_t; \kappa)$; as $\text{GVOL}_t \rightarrow \infty$, the slope increases linearly.¹³

D. Out-of-Sample Results

To compare the different regime-switching specifications in Sections IA.6.A, IA.6.B, and IA.6.C, we generate out-of-sample exchange rate change forecasts based on the different specifications, where the panel predictive regressions include all ten country characteristics as predictors as well as each characteristic interacted with GVOL. Because each panel predictive regression has a relatively large number of predictors (20) and given the inherently large unpredictable component in exchange rate changes, we continue to generate forecasts by estimating the panel predictive regressions via the ENet.

¹³When $z_{i,t} g_t \mathbb{L}(g_t; \kappa)$ is simplified to $z_{i,t} \mathbb{L}(g_t; \kappa)$ in equation (21), the panel predictive regression becomes a conventional logistic smooth transition specification, with the slope coefficient taking the familiar S-shape with a minimum (maximum) value of $\beta^z \frac{1}{\sigma_i^z}$ ($\beta^z \frac{1}{\sigma_i^z} + \beta^{z,g} \frac{1}{\sigma_i^{z,g}}$).

When estimating the interaction with threshold specification in Section IA.6.B, we need to determine a value for the threshold g^* . Recall from Section III.A of the paper that ENet estimation requires tuning the hyperparameters λ and δ , for which we use the ERIC. Our strategy is to treat g^* as an additional hyperparameter to tune; that is, we use the ERIC to tune λ , δ , and g^* jointly for ENet estimation. This has the advantage of setting g^* to improve out-of-sample performance in light of the bias-variance trade-off. We follow a similar strategy for estimating the interaction with LST specification in Section IA.6.C, tuning λ , δ , and the LST curvature parameter κ jointly via the ERIC. To our knowledge, our paper is the first to propose setting the threshold or curvature parameter (g^* or κ , respectively) in the transition function for regime-switching models by treating them as additional hyperparameters to be tuned. Thus, we make a methodological contribution to out-of-sample forecasting with regime-switching specifications for models that are trained using machine-learning methods such as the ENet.

Table IA.16 reports the out-of-sample results. For reference, the fourth column reproduces the results from the sixth column of Table 1 from the paper, which correspond to the panel prediction regression with the ten country characteristics only (i.e., no interaction terms). According to the results in the fifth column of Table IA.16, including interaction terms with GVOL in the panel predictive regression substantially improves out-of-sample performance: the $R^2_{\text{All,OS}}$ statistic goes from negative (-0.29%) to 0.61% (significant at the 5% level), and nearly all of the country-level $R^2_{i,\text{OS}}$ statistics increase (some markedly, e.g., the United Kingdom and New Zealand). The last two columns of Table IA.16 show that augmenting the interaction terms with threshold or LST mechanisms generally produces additional out-of-sample gains. The $R^2_{\text{All,OS}}$ statistics are 1.02% and 0.99% for the interaction with threshold and interaction with LST specifications, respectively (both significant at the 5% level). For countries such as Switzerland,

Australia, New Zealand, Sweden, Norway, Denmark, and the Euro area, the $R_{i,OS}^2$ statistics evince considerable increases in the last two columns.¹⁴ Overall, the results in Table IA.16 highlight the relevance of GVOL for capturing FX market conditions that affect the responses of exchange rate changes to economic fundamentals, especially BILL, reiterating the findings in Sections IV.D and V of the paper.

IA.7. Robustness Checks: Portfolio Performance

A. Allocating Between the US Dollar and an Individual Foreign Currency

We consider a scenario where a mean-variance investor allocates monthly between the US dollar and an individual foreign currency i . At the end of month T , the investor's objective function is given by

$$(23) \quad \max_{w_{i,T+1|T}} w_{i,T+1|T} \widehat{r}x_{i,T+1|T} - 0.5\gamma w_{i,T+1|T}^2 \hat{\sigma}_{i,T+1|T}^2,$$

where $\widehat{r}x_{i,T+1|T}$ is given by equation (14) from the paper, $w_{i,T+1|T}$ is the allocation to currency i that will be in effect for month $T + 1$ (the US dollar allocation is $1 - w_{i,T+1|T}$), and $\hat{\sigma}_{i,T+1|T}^2$ is the investor's estimate of the variance for the currency excess return. We assume that the investor uses an exponentially weighted moving average (EWMA) estimator for $\hat{\sigma}_{i,T+1|T}^2$ (with a decay factor of 0.92). As in the paper, we assume that $\gamma = 5$. To keep the allocation in a range that appears

¹⁴Conventional threshold and LST specifications (footnotes 12 and 13, respectively) generally do not outperform the no-change benchmark forecast. Thus, allowing for interactions appears important for short-run out-of-sample exchange rate prediction.

plausible for an investor allocating to a single asset, we impose the restriction that

$$-1 \leq w_{i,T+1|T} \leq 1.5, \text{ similarly to related studies on stock return predictability.}$$

We consider the two cases from Section VI.A in the paper. In the first case, the Bench-Opt (benchmark optimized) portfolio, the investor sets the expected exchange rate change to the no-change benchmark forecast ($\hat{\Delta}s_{i,T+1|T} = 0$) in equation (14) from the paper. In the second case, the ML-Opt (machine-learning optimized) portfolio, the investor uses the baseline ENet, DNN, or ensemble forecast for the exchange rate change in equation (14) from the paper. To measure the economic value of a machine-learning forecast to the investor, we compute the annualized average utility gain in equation (15) from the paper.

Table IA.17 reports the annualized average utility gain when the investor allocates between the US dollar and an individual foreign currency. The gains accruing to the investor when they use the ML-Opt instead of the Bench-Opt portfolio are all positive and typically quite sizable. The clear majority are above 200 basis points and a number exceed 300 basis points. Table IA.17 demonstrates that the improvements in the statistical accuracy provided by the machine-learning forecasts, as reflected by the $R^2_{i,OS}$ statistics in Table 1 from the paper, also produce substantial economic value from an investment perspective. The results demonstrate the value of the forecasts at the individual currency level in terms of direction (i.e., when to buy and sell a currency) as well as magnitude.

B. Transaction Costs

We examine the effects of transaction costs by computing portfolio excess returns using bid and ask quotes from Datastream for the relevant forward and spot rates (e.g., Lustig et al.

(2011)). In this case, the investor is assumed to pay the bid and ask prices reported in Datastream. A number of studies document that the bid-ask spreads offered by Datastream are unrealistically high (e.g., Lyons (2001), Neely, Weller, and Ulrich (2009), Menkhoff, Sarno, Schmeling, and Schrimpf (2012b), Neely and Weller (2013)). Specifically, investors trade at the best quoted price at each point in time, making the full spread in Datastream considerably higher than the effective spread for FX market participants.¹⁵ To more accurately reflect the relevant transaction costs faced by traders, we follow Goyal and Saretto (2009) and Menkhoff et al. (2012b) and also report results for currency excess returns computed using 25%, 50%, and 75% of the quoted bid-ask spreads from Datastream.

Table IA.19 reports portfolio performance measures adjusted for transaction costs for the 1995:01 to 2020:09 out-of-sample period. As expected, the annualized Sharpe ratios decrease monotonically as we move from Panel A (25% of the bid-ask spread) to Panel D (full bid-ask spread). Nevertheless, they remain quite sizable for the ML-Opt portfolios: they are greater than or equal to 0.69 (0.61) for 25% (50%) of the bid-ask spread in Panel A (B); for 75% of the bid-ask spread in Panel C, they still range from 0.53 (DNN) to 0.73 (baseline ENet).¹⁶ The ML-Opt portfolios also continue to provide substantive economic value to the investor vis-à-vis the Bench-Opt portfolio in terms of the annualized average utility gains in the second column. Even for 75% of the bid-ask spread, the gains range from 170 (DNN) to 424 (baseline ENet) basis

¹⁵The FX market is one of the most liquid markets, with low transaction costs and no natural short-selling constraints. According to the 2016 Bank for International Settlements Triennial Survey, average daily turnover in the FX market is five trillion US dollars.

¹⁶With the exception of the DNN for the full bid-ask spread in Panel D, all of the Sharpe ratios for the ML-Opt portfolios are significant at the 1% level in Panels A through D of Table IA.19.

points. In sum, the performance of the ML-Opt portfolios—and thus the economic value of out-of-sample exchange rate predictability—is robust to reasonable assumptions regarding transaction costs.

C. A Closer Look at Late 2008

Figure 3 from the paper indicates that late 2008 is a perilous time for the Bench-Opt portfolio (as well as the conventional carry portfolio). Figures IA.15 to IA.20 provide additional insight into the sources of the poor performance of the Bench-Opt portfolio in late 2008 as well as how the ML-Opt portfolios improve performance. Figure IA.15 to IA.17 show the currency excess return forecasts for September through December of 2008 for the baseline ENet, DNN, and ensemble log exchange rate change forecasts, respectively, that serve as inputs in the portfolio optimization problem in equation (13) from the paper. The figures also show the currency excess return forecasts for the Bench-Opt portfolio along with the realized excess returns. Because the Bench-Opt portfolio uses the no-change benchmark exchange rate forecast, the benchmark currency excess return forecast in equation (14) from the paper is simply the bill yield differential; the ML-Opt portfolio augments the bill yield differential with the baseline ENet, DNN, or ensemble exchange rate change forecast. Figures IA.18 to IA.20 display the currency weights for the portfolios.

For September of 2008, the machine-learning and benchmark currency excess return forecasts in Figures IA.15 to IA.17 lead to allocations of similar sizes in Figures IA.18 to IA.20 for most countries. The machine-learning currency excess return forecasts generate a notable difference in allocation for the Euro area, where the ML-Opt portfolios (Bench-Opt portfolio)

take short (long) positions; the ML-Opt portfolios based on the DNN and ensemble forecasts take long positions for Japan, while the Bench-Opt portfolio takes a short position. As shown in Figures IA.15 to IA.17, with the exceptions of Japan and Canada, all of the realized currency excess returns are negative for September of 2008; the negative returns are large in magnitude for Australia, New Zealand, Sweden, Norway, Denmark, and the Euro area. On the basis of the allocations and realized currency excess returns, the excess return for the Bench-Opt portfolio is -7.57% in September of 2008, while the losses are smaller for the ML-Opt portfolio, ranging from -4.02% (ensemble) to -5.25% (baseline ENet). The differences in allocations signaled by the machine-learning forecasts—especially for the Euro area—help to limit portfolio losses in the month of the Lehman bankruptcy.

The currency excess return forecasts diverge more sharply during October of 2008 in Figures IA.15 to IA.17: with the exception of Japan, the benchmark currency excess return forecasts are all positive, while the machine-learning forecasts are negative for all ten countries for the baseline ENet forecast and nearly all negative for the DNN and ensemble forecasts. The differences in currency excess return forecasts give rise to a number of markedly different allocations during October of 2008 in Figures IA.18 to IA.20; for example, the ML-Opt (Bench-Opt) portfolios based on the baseline ENet and ensemble forecasts exhibit sizable short (long) positions for the United Kingdom, New Zealand, Norway, Denmark, and the Euro area. With the exception of Japan, all of the realized currency excess returns are negative for October of 2008 in Figures IA.15 to IA.17, and the negative returns are larger in magnitude than those for September of 2008.¹⁷ The different allocations prompted by the machine-learning forecasts

¹⁷Note the difference in scales for the vertical axes across the top two panels of Figures IA.15 to IA.17.

vis-à-vis the no-change benchmark enable the ML-Opt portfolios to substantively outperform the Bench-Opt portfolio during October of 2008: the latter suffers a large loss of -11.60% , while the ML-Opt portfolios realize substantial gains of 6.29% (DNN) to 24.26% (baseline ENet). The situation for November of 2008 is fairly similar to that for October in terms of the currency excess return forecasts and allocations in Figures IA.15 to IA.20. Because the negative realized currency excess returns are typically smaller in magnitude, the Bench-Opt realizes an excess return of 1.91% during the month; however, the information in the machine-learning forecasts leads to much larger excess returns for the ML-Opt portfolios, ranging from 12.36% (ensemble) to 12.90% (DNN).

The environment appears to normalize to an extent in December of 2008, in the sense that the discrepancies between the machine-learning and benchmark forecasts in Figures IA.15 to IA.17, as well as those between the ML-Opt and Bench-Opt portfolio weights in Figures IA.18 to IA.20, are more muted. Nevertheless, important differences remain (e.g., the portfolio weights for Switzerland and Norway). In addition, with the exception of the United Kingdom, all the realized currency excess returns are positive in December of 2008. The Bench-Opt portfolio experiences an excess return of 5.64% for the month, while the excess returns are around three times as large for the ML-Opt portfolios (19.07% , 15.93% , and 17.58% for the baseline ENet, DNN, and ensemble, respectively).¹⁸

Figures IA.15 to IA.20 help to explain how exchange rate predictability—as captured by our machine-learning forecasts—is valuable to an investor during the worst part of the Global

¹⁸For September through December of 2008, the excess returns for the conventional carry trade portfolio are -6.33% , -15.00% , -3.06% , and -4.77% , respectively, in line with the sharp drop in the portfolio's cumulative excess return in Figure 3 from the paper.

Financial Crisis in late 2008. By anticipating depreciations in many foreign currencies, the machine-learning forecasts lead to sizable negative positions in many currencies, enabling the ML-Opt portfolios to avoid the large losses suffered by a Bench-Opt portfolio and even realize substantive gains.

D. Alphas

We examine whether the ML-Opt and Bench-Opt portfolios generate alpha in the context of the Lustig et al. (2011) currency factor model. The model includes dollar and carry trade risk factors, denoted by MKT_{FX} and HML_{FX} , respectively. The dollar factor is an equally weighted average of the available currency excess returns for the month, while the carry trade risk factor is the conventional carry portfolio defined in the paper. Table IA.20 reports factor model estimation results for the ML-Opt and Bench-Opt portfolios. We again report results for the full out-of-sample period as well as the 1995:01 to 2008:08 and 2008:09 to 2020:09 subsamples.

For the full 1995:01 to 2020:09 out-of-sample period in Panel A of Table IA.20, the ML-Opt portfolios generate large annualized alphas of 11.53%, 7.98%, and 9.51% for the baseline ENet, DNN, and ensemble forecasts, respectively, all of which are significant at the 1% level and substantially larger than that for the Bench-Opt portfolio (4.27%). The Bench-Opt portfolio evinces considerable exposure to the carry factor (0.75, significant at the 1% level) for the full out-of-sample period. The ML-Opt portfolios also exhibit sizable exposures to the carry factor (0.40 for all three forecasts, which are significant at the 5% or 1% level), but they are about half as large as that for the Bench-Opt portfolio.

Panels B and C of Table IA.20 reveal important differences in performance for the

Bench-Opt portfolio across the two subsamples. For the 1995:01 to 2008:08 subsample in Panel B, the Bench-Opt portfolio exhibits substantial exposure to the carry trade factor (0.92, significant at the 1% level), and it generates a sizable annualized alpha of 4.51% (significant at the 1% level). Reminiscent of Table 3 and Figure 3 from the paper, the Bench-Opt portfolio's performance deteriorates sharply for the 2008:09 to 2020:09 subsample in Panel C. It continues to display substantial exposure to the carry factor (0.64, significant at the 1% level), while its annualized alpha declines to only 1.71% (insignificant at conventional levels). In contrast, the ML-Opt portfolios deliver impressive annualized alphas for both subsamples in Panels B and C of Table IA.20, all of which are above 585 basis points and significant. An interesting pattern emerges with respect to the exposures of the ML-Opt portfolios to the carry factor. The exposures are quite sizable for the first subsample, ranging from 0.64 (DNN) to 0.76 (baseline ENet), and all are significant at the 1% level. For the second subsample, the exposures become much smaller, ranging from 0.02 (baseline ENet) to 0.19 (DNN), and none is significant. Thus, the information contained in the 70 predictors leads the investor to effectively disconnect fully from a conventional carry strategy for the second subsample. Taken together, the pre- and post-2008 exposures provide direct evidence that the model picks up time-varying risk exposures.

E. Conditional Maximum Sharpe Ratios

Finally, we examine the economic implications of our machine-learning forecasts for the conditional maximum Sharpe ratio (CMSR), which relates to the conditional price of currency risk. Chernov et al. (2023) recently emphasize that it is feasible to construct the ex ante mean-variance efficient portfolio for individual currencies, as the cross section of currencies is

relatively small compared to that for individual equities. The sheer size of the cross section for equities makes estimating the covariance matrix with reasonable precision problematic, so it is practically infeasible to compute a reliable mean-variance efficient portfolio for asset pricing based on individual stocks. In contrast, constructing a reliable covariance matrix estimate is eminently workable for the limited cross section of individual currencies. Combined with a vector of currency excess return forecasts, we can construct an ex ante conditional mean-variance efficient (CMVE) portfolio based on information through month t .

Along the lines of studies such as Ackermann, Pohl, and Schmedders (2017), Daniel, Hodrick, and Lu (2017), Chernov et al. (2023), and Maurer, To, and Tran (2023), we construct a CMVE portfolio by computing the portfolio that is fully invested in the individual foreign currencies and maximizes the expected Sharpe ratio (i.e., the tangency portfolio). Specifically, to estimate the CMSR for month $T + 1$ based on information available through month T , we compute portfolio weights based on the following objective function:

$$(24) \quad \max_{\mathbf{w}_{T+1|T}} \left(\mathbf{w}'_{T+1|T} \hat{\boldsymbol{\mu}}_{T+1|T} \right) \left(\mathbf{w}'_{T+1|T} \hat{\boldsymbol{\Sigma}} \mathbf{w}_{T+1|T} \right)^{-0.5} \quad \text{subject to } \boldsymbol{\iota}'_N \mathbf{w}_{T+1|T} = 1,$$

where $\hat{\boldsymbol{\mu}}_{T+1|T}$ is the vector of currency excess return forecasts, $\hat{\boldsymbol{\Sigma}}$ is the estimate of the covariance matrix for the excess returns, and $\boldsymbol{\iota}_N$ is an N -dimensional vector of ones. To ensure that the portfolio weights lie in reasonable ranges, we also impose the restrictions that $-1 \leq w_{i,T+1|T} \leq 1$ for $i = 1, \dots, N$. Denoting the solution by $\mathbf{w}^*_{T+1|T}$, the CMSR is given by

$$(25) \quad \text{CMSR}_{T+1|T} = \left(\mathbf{w}^{*'}_{T+1|T} \hat{\boldsymbol{\mu}}_{T+1|T} \right) \left(\mathbf{w}^{*'}_{T+1|T} \hat{\boldsymbol{\Sigma}} \mathbf{w}^*_{T+1|T} \right)^{-0.5},$$

which can be viewed as a measure of the conditional price of currency risk. We are interested in how the information in our broad array of predictors—as embodied in the machine-learning exchange rate forecasts—affects the CMSR and price of currency risk. Because our estimates of the CMSR are out-of-sample, they are more relevant for investors in real time.

Figure IA.21 plots the annualized CMSR when $\hat{\mu}_{T+1|T}$ uses the baseline ENet, DNN, ensemble and no-change benchmark forecasts of the exchange rate change (considered in turn) when forming the currency excess return forecasts. As in Section VI of the paper, $\hat{\Sigma}$ is based on an EWMA estimator. The CMSRs based on the machine-learning forecasts move together fairly closely over time, ranging from approximately 0.3 to over four. They indicate an economically sizable CMSR as well as economically significant fluctuations in the conditional price of currency risk. Periods with an elevated CMSR based on the machine-learning forecasts are often associated with times of stress in the global economy. For example, the CMSRs evince marked increases in the mid-to-late 2000s during the Global Financial Crisis and concomitant Great Recession. Sharp increases are also evident in early 2020 corresponding to the advent of the COVID-19 pandemic. Compared to those based on the machine-learning forecasts, the CMSR based on the no-change benchmark forecast tends to be lower and less volatile. Overall, our approach for forecasting exchange rates, which employs a large information set and machine learning, yields a sensibly behaved conditional price of currency risk, with notable increases during periods of global economic stress.

F. Portfolio Performance and Global FX Volatility

In Section IV.D of the paper, we examine the relation between global FX volatility and the statistical accuracy of the machine-learning forecasts relative to the no-change benchmark. The results in Table 2 from the paper provide extensive evidence that the relative accuracy of the machine-learning forecasts significantly improves as global FX volatility increases. Analogously, we investigate the relation between global FX volatility and the economic value of the machine-learning forecasts vis-à-vis the no-change benchmark. To do so, equation (11) from the paper becomes

$$(26) \quad rx_{i,t}^{\text{ML-Opt}} - rx_{i,t}^{\text{Bench-Opt}} = a_{i,0} + a_{i,1}\text{GVOL}_t + \varepsilon_{i,t},$$

where $rx_{i,t}^{\text{ML-Opt}}$ ($rx_{i,t}^{\text{Bench-Opt}}$) is the month- t excess return for the mean-variance US investor who allocates monthly between the US dollar and the foreign currency for country i using a machine-learning exchange rate change (no-change benchmark) forecast, corresponding to the objective function in equation (23). We also estimate a version of equation (26) for the scenario where the US investor allocates monthly across the US dollar and all available foreign currencies, corresponding to the objective function in equation (13) from the paper:

$$(27) \quad rx_t^{\text{ML-Opt}} - rx_t^{\text{Bench-Opt}} = a_0 + a_1\text{GVOL}_t + \varepsilon_t,$$

where $rx_t^{\text{ML-Opt}}$ ($rx_t^{\text{Bench-Opt}}$) is the month- t excess return for the mean-variance portfolio based on a machine-learning exchange rate change (no-change benchmark) forecast. To facilitate the interpretation of the results, we standardize GVOL_t in equations (26) and (27).

Table IA.21 reports the results. With the exception of Japan for the baseline ENet forecast, all of the slope coefficient estimates are positive and significant (nearly all at the 1% level). Thus, we have strong evidence that the performance of the ML-Opt portfolios significantly improves relative to the Bench-Opt portfolios as GVOL increases, analogously to the extensive evidence in Table 2 from the paper that the statistical accuracy of the machine-learning forecasts significantly improves relative to the no-change benchmark as GVOL increases. The magnitudes of the slope coefficient estimates are also economically sizable. For example, for the portfolios that invest in all available foreign currencies and the baseline ENet forecast, a one-standard-deviation increase in GVOL is associated on average with a 131 basis point increase in the monthly excess return difference between the ML-Opt and Bench-Opt portfolios. Many of the other slope coefficients are around 50 basis points or larger, again signaling quite large increases in excess return differences on average.

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TABLE IA.1

Summary Statistics

The table reports summary statistics for monthly log exchange rate changes measured against the US dollar. The country- i log exchange rate change is $\Delta s_{i,t}$, where $s_{i,t} = \log(S_{i,t})$, and $S_{i,t}$ is the month- t spot exchange rate for country i (number of country- i currency units per US dollar). The annualized mean (volatility) in the third (fourth) column is the monthly mean (standard deviation) multiplied by 12 ($\sqrt{12}$).

1	2	3	4	5	6	7
Country	Sample period	Ann. mean	Ann. vol.	Skewness	Excess kurtosis	Autocorr.
United Kingdom	1985:01–2020:09	−0.30%	9.98%	0.30	2.52	0.06
Switzerland	1985:01–2020:09	−2.90%	11.14%	−0.04	0.96	−0.01
Japan	1985:01–2020:09	−2.43%	10.88%	−0.35	1.82	0.04
Canada	1985:01–2020:09	0.02%	7.34%	0.48	3.99	−0.04
Australia	1985:01–2020:09	0.39%	11.64%	0.67	2.36	0.04
New Zealand	1985:01–2020:09	−0.92%	12.04%	0.39	1.83	−0.03
Sweden	1985:01–2020:09	−0.01%	10.98%	0.44	1.50	0.10
Norway	1985:01–2020:09	0.08%	11.03%	0.43	1.11	0.02
Denmark	1985:01–2020:09	−1.61%	10.27%	0.19	0.79	0.03
Euro area	1999:02–2020:09	−0.14%	9.55%	0.16	1.13	0.03
Germany	1985:01–1998:12	−4.55%	11.61%	0.32	0.32	0.03
Italy	1985:01–1998:12	−1.13%	11.39%	0.79	2.03	0.09
France	1985:01–1998:12	−3.89%	11.13%	0.42	0.54	0.01
Netherlands	1985:01–1998:12	−4.56%	11.57%	0.28	0.34	0.03

TABLE IA.2

Publication Lags: CPIs and Unemployment Rates

The table reports publication frequencies and lags for CPIs and unemployment rates for non-US countries. The last column shows the adjustment made to the timing of the country characteristic to account for the publication lag. Adjustments for the variables reported at a quarterly frequency are described in Section IA.1.B.

1	2	3	4
Country	Publication frequency	Publication lag	Adjustment
<i>Panel A. CPI</i>			
United Kingdom	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Switzerland	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Japan	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Canada	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Australia	Quarterly	—	—
New Zealand	Quarterly	—	—
Sweden	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Norway	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Denmark	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Euro area	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Germany	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Italy	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
France	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$
Netherlands	Monthly	1 month	$INF_{i,t} \Rightarrow INF_{i,t-1}$

TABLE IA.2 (continued)

1	2	3	4
Country	Publication frequency	Publication lag	Adjustment
<i>Panel B. Unemployment rate</i>			
United Kingdom	Monthly	2 months	$UN_{i,t} \Rightarrow UN_{i,t-2}$
Switzerland	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$
Japan	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$
Canada	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$
Australia	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$
New Zealand	Quarterly	—	—
Sweden	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$
Norway	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$
Denmark	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$
Euro area	Monthly	2 months	$UN_{i,t} \Rightarrow UN_{i,t-2}$
Germany	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$
Italy	Monthly	2 months	$UN_{i,t} \Rightarrow UN_{i,t-2}$
France	Monthly	2 months	$UN_{i,t} \Rightarrow UN_{i,t-2}$
Netherlands	Monthly	1 month	$UN_{i,t} \Rightarrow UN_{i,t-1}$

TABLE IA.3

Correlation Matrix: Global State Variables

The table reports the correlation matrix for the global state variables defined in Section II of the paper; 0.00 indicates less than 0.005 in absolute value. The sample period is 1985:01 to 2020:09.

1	2	3	4	5	6	7
Global variable	EPU	MPU	GR	GVOL	GILL	GCOR
EPU	1	0.50	0.13	0.19	0.00	0.04
MPU	0.50	1	0.40	0.26	0.12	0.04
GR	0.13	0.40	1	−0.02	−0.03	0.06
GVOL	0.19	0.26	−0.02	1	0.33	0.27
GILL	0.00	0.12	−0.03	0.33	1	−0.03
GCOR	0.04	0.04	0.06	0.27	−0.03	1

TABLE IA.4

Out-of-Sample R^2 Statistics: Countries that Joined the Euro

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. “Baseline OLS” (“Baseline ENet”) is the baseline panel predictive regression in equation (1) from the paper estimated via OLS (the ENet); “No-interaction ENet” is the panel predictive regression without interaction terms in equation (3) from the paper estimated via the ENet; “DNN” is the deep neural network in Section III.B of the paper; “Ensemble” is an average of the baseline ENet and DNN forecasts.

1	2	3	4	5	6	7	8
Country	Out-of-sample period	Obs.	Baseline OLS	Baseline ENet	No-inter- action ENet	DNN	Ensemble
Germany	1995:01–1998:12	48	−9.77	0.38	1.16	−3.86	−1.65
Italy	1995:01–1998:12	48	−9.90	0.18	1.05	0.07	0.26
France	1995:01–1998:12	48	−32.97	0.68	0.74	−1.39	−0.22
Netherlands	1995:01–1998:12	48	−19.34	0.52	2.30	−3.50	−1.24

TABLE IA.5

Out-of-Sample R^2 Statistics: Japanese Yen as Base Currency

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes with the Japanese yen serving as the base currency. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. “Baseline OLS” (“Baseline ENet”) is the baseline panel predictive regression in equation (1) from the paper estimated via OLS (the ENet); “DNN” is the deep neural network in Section III.B of the paper; “Ensemble” is an average of the baseline ENet and DNN forecasts.

1	2	3	4	5	6	7
Country	Out-of-sample period	Observations	Baseline OLS	Baseline ENet	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	−4.91	1.94**	0.85	1.67**
Switzerland	1995:01–2020:09	309	−10.77	0.51	2.02	1.65
United States	1995:01–2020:09	309	−13.35	−1.71	−1.28	−1.28
Canada	1995:01–2020:09	309	−6.03	1.15*	0.54	1.09*
Australia	1995:01–2020:09	309	−1.41	1.31*	−0.33	0.69
New Zealand	1995:01–2020:09	309	−5.10	1.68*	0.89	1.54*
Sweden	1995:01–2020:09	309	1.82**	1.66*	1.38*	1.73*
Norway	1995:01–2020:09	309	−3.67	1.97*	0.81	1.65*
Denmark	1995:01–2020:09	309	−3.74	1.57*	0.62	1.37*
Euro area	2000:02–2020:09	248	−7.61	1.82	−3.01	−0.06
All	1995:01–2020:09	3,029	−4.84	1.31***	0.39***	1.12***

TABLE IA.6

Out-of-Sample R^2 Statistics: Swiss Franc as Base Currency

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes with the Swiss franc serving as the base currency. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. “Baseline OLS” (“Baseline ENet”) is the baseline panel predictive regression in equation (1) from the paper estimated via OLS (the ENet); “DNN” is the deep neural network in Section III.B of the paper; “Ensemble” is an average of the baseline ENet and DNN forecasts.

1	2	3	4	5	6	7
Country	Out-of-sample period	Observations	Baseline OLS	Baseline ENet	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	−21.57	0.67*	0.31	0.93*
United States	1995:01–2020:09	309	−21.70	−3.69	−1.50	−1.50
Japan	1995:01–2020:09	309	−13.17	−1.48	−1.40	−1.17
Canada	1995:01–2020:09	309	−14.29	−0.63	−0.93	−0.39
Australia	1995:01–2020:09	309	−9.74	0.11	−1.41	−0.34
New Zealand	1995:01–2020:09	309	−21.72	0.12	−1.33	−0.17
Sweden	1995:01–2020:09	309	−15.97	−0.23	−0.60	0.01
Norway	1995:01–2020:09	309	−17.27	0.91*	−0.05	0.98*
Denmark	1995:01–2020:09	309	−48.26	−5.89	−2.32	−3.08
Euro area	2000:02–2020:09	248	−51.52	−6.97	−3.49	−3.67
All	1995:01–2020:09	3,029	−18.98	−1.01	−1.10	−0.52***

TABLE IA.7

Out-of-Sample R^2 Statistics: Different Hyperparameter Tuning Methods

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. The fourth through seventh columns report results for baseline ENet forecasts that use five-fold cross validation (5-fold CV), time-series validation (TSV), time-series cross validation (TSCV), and the BIC, respectively, to tune the regularization parameters (δ and λ), as described in IA.4.A.

1	2	3	4	5	6	7
Country	Out-of-sample period	Observations	5-fold CV	TSV	TSCV	BIC
United Kingdom	1995:01–2020:09	309	−3.99	1.75*	2.46**	3.35**
Switzerland	1995:01–2020:09	309	−5.38	−1.02	0.96**	0.13*
Japan	1995:01–2020:09	309	−6.37	−0.98	0.50*	−0.32
Canada	1995:01–2020:09	309	−10.81	−3.57	−1.26	−2.77
Australia	1995:01–2020:09	309	−0.98	0.33*	−0.15	1.07**
New Zealand	1995:01–2020:09	309	−3.83	−1.60	1.16	2.81**
Sweden	1995:01–2020:09	309	−0.48	2.20**	1.87**	1.62**
Norway	1995:01–2020:09	309	−1.25	1.98**	1.43**	1.57**
Denmark	1995:01–2020:09	309	−0.85	0.86***	0.85***	2.23***
Euro area	2000:02–2020:09	248	−5.07	−0.20	−0.88	1.23**
All	1995:01–2020:09	3,029	−3.51	0.02***	0.76***	1.22***

TABLE IA.8

Out-of-Sample R^2 Statistics: Random Features Forecast

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes based on the Kelly et al. (2024) random features model. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. P is the number of random features.

1	2	3	4	5	6
Country	Out-of-sample period	Observations	$P = 10,000$	$P = 20,000$	$P = 30,000$
United Kingdom	1995:01–2020:09	309	1.58*	1.58**	1.40*
Switzerland	1995:01–2020:09	309	0.57	0.07	0.05
Japan	1995:01–2020:09	309	−1.69	−1.29	−1.55
Canada	1995:01–2020:09	309	−2.79	−2.14	−2.98
Australia	1995:01–2020:09	309	−0.24	0.48**	−0.10
New Zealand	1995:01–2020:09	309	2.05**	2.08**	1.94**
Sweden	1995:01–2020:09	309	1.23*	1.59*	1.50*
Norway	1995:01–2020:09	309	−0.57	0.12*	−0.15
Denmark	1995:01–2020:09	309	−1.05	−0.44	−0.72
Euro area	2000:02–2020:09	248	−4.68	−2.66	−4.14
All	1995:01–2020:09	3,029	−0.39	0.05**	−0.30

TABLE IA.9

Out-of-Sample R^2 Statistics: Ridge-Based Forecasts

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. “Baseline ridge” is the baseline predictive regression in equation (1) from the paper estimated via ridge regression; “DNN” is the deep neural network in Section III.B of the paper; “Ensemble” is an average of the baseline ridge and DNN forecasts.

1	2	3	4	5	6
Country	Out-of-sample period	Observations	Baseline ridge	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	0.30**	3.03**	1.99**
Switzerland	1995:01–2020:09	309	0.15*	2.15**	1.37**
Japan	1995:01–2020:09	309	0.09*	0.28	0.35
Canada	1995:01–2020:09	309	0.08	0.91	0.80
Australia	1995:01–2020:09	309	0.05	1.89***	1.17**
New Zealand	1995:01–2020:09	309	0.20**	1.96**	1.25**
Sweden	1995:01–2020:09	309	0.15*	1.69**	1.10**
Norway	1995:01–2020:09	309	0.15*	1.55**	1.06**
Denmark	1995:01–2020:09	309	0.15**	2.04**	1.30***
Euro area	2000:02–2020:09	248	0.29*	2.11**	1.48**
All	1995:01–2020:09	3,029	0.15**	1.74***	1.16***

TABLE IA.10

Out-of-Sample R^2 Statistics: Models Fitted Using Country-Specific Data

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. “Baseline OLS” (“Baseline ENet”) is the baseline predictive regression in equation (1) from the paper estimated via OLS (the ENet); “DNN” is the deep neural network in Section III.B of the paper; “Ensemble” is an average of the baseline ENet and DNN forecasts. The models are estimated using country-specific data.

1	2	3	4	5	6	7
Country	Out-of-sample period	Observations	Baseline OLS	Baseline ENet	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	−161.92	0.00	0.83**	0.48**
Switzerland	1995:01–2020:09	309	−193.02	0.00	−0.07	0.13
Japan	1995:01–2020:09	309	−170.17	0.00	0.51	0.33
Canada	1995:01–2020:09	309	−65.94	0.42	0.34	0.47
Australia	1995:01–2020:09	309	−104.08	0.50**	1.26**	1.12***
New Zealand	1995:01–2020:09	309	−134.92	1.35***	−0.11	0.72**
Sweden	1995:01–2020:09	309	−79.12	0.00	0.51	0.36
Norway	1995:01–2020:09	309	−93.55	0.00	0.74*	0.44*
Denmark	1995:01–2020:09	309	−120.32	0.00	0.16	0.20

TABLE IA.11

Out-of-Sample R^2 Statistics: Random Forest

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. The ensemble-all forecast is an average of the baseline ENet, DNN, and random forest forecasts.

1	2	3	4	5
Country	Out-of-sample period	Observations	Random forest	Ensemble-all
United Kingdom	1995:01–2020:09	309	2.14**	3.72**
Switzerland	1995:01–2020:09	309	1.23**	2.03**
Japan	1995:01–2020:09	309	−1.11	0.33
Canada	1995:01–2020:09	309	−1.60	0.73*
Australia	1995:01–2020:09	309	0.62**	1.85**
New Zealand	1995:01–2020:09	309	2.14***	2.54***
Sweden	1995:01–2020:09	309	1.99**	2.46**
Norway	1995:01–2020:09	309	2.23***	2.49***
Denmark	1995:01–2020:09	309	1.49***	2.64***
Euro area	2000:02–2020:09	248	0.76**	2.66**
All	1995:01–2020:09	3,029	1.10***	2.13***

TABLE IA.12

Granger Causality Test Results

Panel A reports Granger causality test results for monthly log exchange rate changes (LIRET) and net commodity returns (CXP) at the country level, with the lag order selected via the AIC (considering a maximum lag of twelve). Panel B provides robustness checks across fixed lag orders, reporting the number and percent of countries for which the null of no Granger causality is rejected at the 10% level. Panel C reports results for the Dumitrescu and Hurlin (2012) panel Granger causality test based on the $\bar{Z}$ statistic; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively; 0.00 indicates less than 0.005.

1	2	3	4	5	6	7
			LIRET $\rightarrow$ CXP		CXP $\rightarrow$ LIRET	
<i>Panel A. Country-level tests (AIC lag selection)</i>						
Country	Obs.	Lag	<i>F</i> -stat	<i>p</i> -value	<i>F</i> -stat	<i>p</i> -value
United Kingdom	443	7	1.14	0.33	0.93	0.49
Switzerland	443	1	14.17	0.00***	0.96	0.33
Japan	443	1	0.01	0.91	0.09	0.77
Canada	429	1	31.73	0.00***	4.18	0.04**
Australia	429	5	9.91	0.00***	1.39	0.23
New Zealand	429	6	2.44	0.03**	0.41	0.87
Sweden	429	2	0.69	0.51	0.39	0.67
Norway	429	1	19.72	0.00***	2.58	0.11
Denmark	429	1	2.11	0.15	0.58	0.45
Euro area	260	1	9.56	0.00***	0.30	0.58
Germany	182	1	3.08	0.08*	0.78	0.38
Italy	182	1	4.26	0.04**	0.51	0.48
France	182	1	2.88	0.09*	2.89	0.09*
Netherlands	182	1	0.62	0.43	0.69	0.41

TABLE IA.12 (continued)

1	2	3	4	5	6	7
LIRET $\rightarrow$ CXP				CXP $\rightarrow$ LIRET		
<i>Panel B: Robustness checks (fixed lag orders)</i>						
Lag	N	# significant	% significant	# significant	% significant	
1	14	8	57	2	14	
3	14	8	57	0	0	
6	14	7	50	1	7	
12	14	6	43	2	14	
<i>Panel C: Panel tests</i>						
Lag	N	$\bar{Z}$	p -value	$\bar{Z}$	p -value	
1	14	19.70	0.00***	0.58	0.28	
3	14	12.83	0.00***	-0.51	0.70	
6	14	10.42	0.00***	0.73	0.23	
12	14	6.55	0.00***	-0.08	0.53	

TABLE IA.13

Out-of-Sample R^2 Statistics: Including Net Commodity Returns

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes with net commodity returns as an additional characteristic. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. “Baseline OLS” (“Baseline ENet”) is the baseline panel predictive regression in equation (1) from the paper estimated via OLS (the ENet); “DNN” is the deep neural network in Section III.B from the paper; “Ensemble” is an average of the baseline ENet and DNN forecasts.

1	2	3	4	5	6	7
Country	Out-of-sample period	Observations	Baseline OLS	Baseline ENet	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	−14.13	2.32**	1.77**	2.33**
Switzerland	1995:01–2020:09	309	−14.20	0.31	0.16	0.59
Japan	1995:01–2020:09	309	−12.36	−1.13	−0.42	−0.53
Canada	1995:01–2020:09	309	−19.75	−0.07	0.09	0.49
Australia	1995:01–2020:09	309	−3.97	0.44	1.39**	1.12**
New Zealand	1995:01–2020:09	309	−8.22	2.39**	1.12**	1.94***
Sweden	1995:01–2020:09	309	−3.35	1.33*	1.13**	1.40**
Norway	1995:01–2020:09	309	−3.21	1.52**	1.24**	1.63**
Denmark	1995:01–2020:09	309	−5.78	1.68***	1.37***	1.77***
Euro area	2000:02–2020:09	248	−11.21	1.82*	1.23**	2.09**
All	1995:01–2020:09	3,029	−8.74	1.07**	0.92***	1.27***

TABLE IA.14

Out-of-Sample R^2 Statistics: Benchmark based on INF and UN

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the benchmark forecast; for the positive R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. The benchmark forecast is based on a linear panel predictive regression with INF and UN as predictors estimated via OLS. “Baseline ENet” is the baseline predictive regression in equation (1) from the paper estimated via the ENet; “DNN” is the deep neural network in Section III.B from the paper; “Ensemble” is an average of the baseline ENet and DNN forecasts.

1	2	3	4	5	6
Country	Out-of-sample period	Observations	Baseline ENet	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	3.33**	3.12**	3.51**
Switzerland	1995:01–2020:09	309	0.09	1.88**	1.32*
Japan	1995:01–2020:09	309	−0.84	−0.18	−0.27
Canada	1995:01–2020:09	309	−0.91	0.01	−0.03
Australia	1995:01–2020:09	309	0.70	1.12*	1.05
New Zealand	1995:01–2020:09	309	1.82*	1.63*	1.86*
Sweden	1995:01–2020:09	309	1.53*	1.36*	1.62*
Norway	1995:01–2020:09	309	1.70**	1.34*	1.72**
Denmark	1995:01–2020:09	309	2.36**	2.18**	2.48**
Euro area	2000:02–2020:09	248	2.47*	2.21**	2.75**
All	1995:01–2020:09	3,029	1.19***	1.42***	1.54***

TABLE IA.15

Out-of-Sample R^2 Statistics: Benchmark based on INF and BILL

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the benchmark forecast; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. The benchmark forecast is based on a linear panel predictive regression with INF and BILL as predictors estimated via OLS. “Baseline ENet” is the baseline predictive regression in equation (1) from the paper estimated via the ENet; “DNN” is the deep neural network in Section III.B in the paper; “Ensemble” is an average of the baseline ENet and DNN forecasts.

1	2	3	4	5	6
Country	Out-of-sample period	Observations	Baseline ENet	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	3.88**	3.67**	4.05**
Switzerland	1995:01–2020:09	309	0.23	2.02**	1.46*
Japan	1995:01–2020:09	309	−0.18	0.47	0.39
Canada	1995:01–2020:09	309	0.44	1.35*	1.31*
Australia	1995:01–2020:09	309	1.71*	2.13***	2.07**
New Zealand	1995:01–2020:09	309	2.63**	2.45**	2.68**
Sweden	1995:01–2020:09	309	2.07**	1.90**	2.16**
Norway	1995:01–2020:09	309	2.21**	1.85**	2.23**
Denmark	1995:01–2020:09	309	2.95***	2.77***	3.07***
Euro area	2000:02–2020:09	248	2.81*	2.55**	3.09**
All	1995:01–2020:09	3,029	1.85***	2.07***	2.20***

TABLE IA.16

Out-of-Sample R^2 Statistics: Regime-Switching Specifications

The table reports out-of-sample R^2 statistics in percent for forecasts of monthly log exchange rate changes. The out-of-sample R^2 statistic measures the proportional reduction in MSPE for the competing forecast in the column heading vis-à-vis the no-change benchmark forecast; 0.00 indicates less than 0.005 in absolute value; for the positive out-of-sample R^2 statistics, *, **, and *** indicate that the reduction in MSPE is significant at the 10%, 5%, and 1% levels, respectively, according to the Clark and West (2007) test. “No interaction” is a panel predictive regression based on ten country characteristics; “Interaction” is a panel predictive regression based on ten country characteristics and their interactions with global FX volatility; “Interaction with threshold” (“Interaction with LST”) augments the interaction terms with a threshold (logistic smooth transition, LST) mechanism. The panel predictive regressions are estimated via the ENet.

1	2	3	4	5	6	7
Country	Out-of-sample period	Obs.	No interaction	Interaction	Interaction with threshold	Interaction with LST
United Kingdom	1995:01–2020:09	309	0.28**	3.04**	2.08*	2.77**
Switzerland	1995:01–2020:09	309	−0.81	−0.20	−0.28	0.17*
Japan	1995:01–2020:09	309	−2.01	−0.83	−1.88	−1.32
Canada	1995:01–2020:09	309	−3.10	−2.95	−2.77	−3.70
Australia	1995:01–2020:09	309	0.77*	0.57*	1.97**	1.14**
New Zealand	1995:01–2020:09	309	0.21*	2.74**	3.58**	3.48**
Sweden	1995:01–2020:09	309	−0.20	0.49	1.65**	1.66**
Norway	1995:01–2020:09	309	0.20*	−0.01	0.23*	0.67**
Denmark	1995:01–2020:09	309	1.02**	1.08**	1.61***	1.81***
Euro area	2000:02–2020:09	248	−0.50	1.11*	2.22**	1.27**
All	1995:01–2020:09	3,029	−0.29	0.61**	1.02**	0.99**

TABLE IA.17

Annualized Average Utility Gains: Investing in Individual Foreign Currencies

The table reports annualized average utility gains in percent for a mean-variance US investor with a relative risk aversion coefficient of five who allocates monthly between the US dollar and the foreign currency in the first column. The average utility gain is the increase in certainty equivalent return when the investor uses the ML-Opt instead of the Bench-Opt portfolio. For the ML-Opt (Bench-Opt) portfolio, the investor uses the baseline ENet, DNN, or ensemble (no-change) exchange rate forecast when predicting the currency excess return.

1	2	3	4	5	6
Country	Out-of-sample period	Obs.	Baseline ENet	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	3.48	2.20	3.16
Switzerland	1995:01–2020:09	309	1.61	1.50	1.86
Japan	1995:01–2020:09	309	0.61	1.55	1.09
Canada	1995:01–2020:09	309	2.40	1.41	1.74
Australia	1995:01–2020:09	309	2.78	1.88	2.27
New Zealand	1995:01–2020:09	309	2.69	2.83	2.58
Sweden	1995:01–2020:09	309	3.16	1.93	2.71
Norway	1995:01–2020:09	309	2.42	2.06	2.38
Denmark	1995:01–2020:09	309	3.25	2.58	3.15
Euro area	2000:02–2020:09	248	3.84	2.82	3.64

TABLE IA.18

Portfolio Performance for Non-US Domestic Investors

The table reports portfolio performance metrics for a mean-variance non-US domestic investor (given in the panel heading) with a relative risk aversion coefficient of five who allocates monthly across the domestic currency and all available foreign currencies. For the ML-Opt (Bench-Opt) portfolio, the investor uses the baseline ENet, DNN, or ensemble (no-change) exchange rate forecast when predicting the currency excess return. The second column reports the annualized increase in certainty equivalent return when the investor uses the ML-Opt instead of the Bench-Opt portfolio. Statistical significance for the Sharpe ratio is based on the Bao (2009) procedure; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

1	2	3	4	5
Portfolio	Annualized average utility gain	Annualized mean	Annualized volatility	Annualized Sharpe ratio
<i>Panel A: United Kingdom</i>				
ML-Opt, baseline ENet	2.92%	10.50%	12.96%	0.81***
ML-Opt, DNN	1.29%	8.40%	12.23%	0.67***
ML-Opt, ensemble	2.43%	9.97%	12.42%	0.78***
Bench-Opt	—	6.19%	10.61%	0.58***
<i>Panel B: Switzerland</i>				
ML-Opt, baseline ENet	−0.49%	7.83%	14.74%	0.53***
ML-Opt, DNN	−1.43%	6.05%	13.54%	0.45**
ML-Opt, ensemble	−0.94%	7.13%	14.39%	0.50**
Bench-Opt	—	6.61%	12.19%	0.54***
<i>Panel C: Japan</i>				
ML-Opt, baseline ENet	0.48%	9.74%	14.41%	0.68***
ML-Opt, DNN	0.82%	8.97%	12.76%	0.70***
ML-Opt, ensemble	0.74%	9.37%	13.51%	0.69***
Bench-Opt	—	7.61%	11.88%	0.64***

TABLE IA.18 (continued)

1	2	3	4	5
Portfolio	Annualized average utility gain	Annualized mean	Annualized volatility	Annualized Sharpe ratio
Panel D: Canada				
ML-Opt, baseline ENet	2.29%	9.34%	12.57%	0.74***
ML-Opt, DNN	0.82%	7.41%	11.84%	0.63***
ML-Opt, ensemble	1.18%	8.07%	12.33%	0.65***
Bench-Opt	—	5.86%	10.54%	0.56***
Panel E: Australia				
ML-Opt, baseline ENet	1.51%	9.52%	12.44%	0.77***
ML-Opt, DNN	0.91%	8.47%	11.68%	0.72***
ML-Opt, ensemble	1.24%	9.08%	12.17%	0.75***
Bench-Opt	—	7.04%	10.77%	0.65***

TABLE IA.19

Portfolio Performance with Transaction Costs

The table reports portfolio performance metrics for a mean-variance US investor with a relative risk aversion coefficient of five who allocates monthly across the US dollar and all available foreign currencies for the 1995:01 to 2020:09 out-of-sample period. Results are reported for different assumptions regarding transaction costs for bid-ask spreads from Datastream. For the ML-Opt (Bench-Opt) portfolio, the investor uses the baseline ENet, DNN, or ensemble (no-change) exchange rate forecast when predicting the currency excess return. The second column reports the annualized increase in certainty equivalent return when the investor uses the ML-Opt instead of the Bench-Opt portfolio. Statistical significance for the Sharpe ratio is based on the Bao (2009) procedure; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

1	2	3	4	5
Portfolio	Annualized average utility gain	Annualized mean	Annualized volatility	Annualized Sharpe ratio
<i>Panel A: 25% of bid-ask spread</i>				
ML-Opt, baseline ENet	4.40%	11.85%	13.53%	0.88***
ML-Opt, DNN	1.82%	8.06%	11.61%	0.69***
ML-Opt, ensemble	2.86%	9.69%	12.59%	0.77***
Bench-Opt	—	5.37%	10.00%	0.54***
<i>Panel B: 50% of bid-ask spread</i>				
ML-Opt, baseline ENet	4.32%	10.86%	13.53%	0.80***
ML-Opt, DNN	1.76%	7.10%	11.60%	0.61***
ML-Opt, ensemble	2.79%	8.71%	12.58%	0.69***
Bench-Opt	—	4.46%	9.99%	0.45**

TABLE IA.19 (continued)

1	2	3	4	5
Portfolio	Annualized average utility gain	Annualized mean	Annualized volatility	Annualized Sharpe ratio
<i>Panel C: 75% of bid-ask spread</i>				
ML-Opt, baseline ENet	4.24%	9.88%	13.52%	0.73***
ML-Opt, DNN	1.70%	6.14%	11.60%	0.53***
ML-Opt, ensemble	2.72%	7.74%	12.57%	0.62***
Bench-Opt	–	3.57%	9.99%	0.36*
<i>Panel D: Full bid-ask spread</i>				
ML-Opt, baseline ENet	4.16%	8.90%	13.52%	0.66***
ML-Opt, DNN	1.65%	5.18%	11.59%	0.45**
ML-Opt, ensemble	2.65%	6.77%	12.57%	0.54***
Bench-Opt	–	2.66%	9.99%	0.27

TABLE IA.20

Alphas

The table reports Lustig et al. (2011) factor model estimation results for ML-Opt portfolios based on the baseline ENet, DNN, and ensemble forecasts as well as the Bench-Opt portfolio. For the ML-Opt (Bench-Opt) portfolios, a mean-variance US investor with a relative risk aversion coefficient of five allocates monthly across available foreign currencies using the baseline ENet, DNN, or ensemble (no-change) exchange rate forecasts when predicting the currency excess return. For the second through fourth columns, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

1	2	3	4	5
Portfolio	Annualized alpha	MKT _{FX}	HML _{FX}	Adjusted R^2
<i>Panel A: 1995:01 to 2020:09 out-of-sample period</i>				
ML-Opt, baseline ENet	11.53%***	−0.47***	0.40**	10.96%
ML-Opt, DNN	7.98%***	−0.06	0.40***	10.06%
ML-Opt, ensemble	9.51%***	−0.28**	0.40**	9.15%
Bench-Opt	4.04%***	−0.03	0.75***	52.81%
<i>Panel B: 1995:01 to 2008:08 out-of-sample period</i>				
ML-Opt, baseline ENet	8.70%***	−0.44	0.76***	32.09%
ML-Opt, DNN	7.94%***	0.09	0.64***	24.84%
ML-Opt, ensemble	7.75%***	−0.16	0.74***	30.87%
Bench-Opt	4.51%***	0.18	0.92***	63.49%
<i>Panel C: 2008:09 to 2020:09 out-of-sample period</i>				
ML-Opt, baseline ENet	12.00%**	−0.25	0.02	0.80%
ML-Opt, DNN	5.85%*	−0.05	0.19	1.11%
ML-Opt, ensemble	8.67%*	−0.16	0.08	−0.59%
Bench-Opt	1.71%	−0.12	0.64***	45.26%

TABLE IA.21

Global FX Volatility and Portfolio Return Differences

The table reports OLS estimates of $a_{i,1}$ for the regression,

$$rx_{i,t}^{\text{ML-Opt}} - rx_{i,t}^{\text{Bench-Opt}} = a_{i,0} + a_{i,1}\text{GVOL}_t + \varepsilon_{i,t}, \text{ where } rx_{i,t}^{\text{ML-Opt}} (rx_{i,t}^{\text{Bench-Opt}}) \text{ is the excess return}$$

for a mean-variance US investor with a relative risk aversion coefficient of five who allocates monthly between the US dollar and the foreign currency for country i in the first column using the machine-learning exchange rate change forecast in the column heading (no-change benchmark forecast), and GVOL_t is global FX volatility. “All” corresponds to the regression,

$$rx_t^{\text{ML-Opt}} - rx_t^{\text{Bench-Opt}} = a_0 + a_1\text{GVOL}_t + \varepsilon_t, \text{ where the US investor allocates monthly across the}$$

US dollar and all available foreign currencies. The excess return differences are measured in

percent, and GVOL_t is standardized; *, **, and *** indicate significance at the 10%, 5%, and 1%

levels, respectively.

1	2	3	4	5	6
Country	Out-of-sample period	Obs.	Baseline ENet	DNN	Ensemble
United Kingdom	1995:01–2020:09	309	0.54***	0.28***	0.51***
Switzerland	1995:01–2020:09	309	0.29***	0.27***	0.34***
Japan	1995:01–2020:09	309	−0.07	0.32***	0.12*
Canada	1995:01–2020:09	309	0.64***	0.39***	0.58***
Australia	1995:01–2020:09	309	0.54***	0.22***	0.38***
New Zealand	1995:01–2020:09	309	0.49***	0.22***	0.44***
Sweden	1995:01–2020:09	309	0.54***	0.19***	0.42***
Norway	1995:01–2020:09	309	0.77***	0.38***	0.66***
Denmark	1995:01–2020:09	309	0.51***	0.34***	0.54***
Euro area	2000:02–2020:09	248	0.99***	0.90***	0.98***
All	1995:01–2020:09	3,029	1.31***	0.85***	1.19***

FIGURE IA.1

Baseline OLS Forecasts

Each panel depicts the realized value and baseline OLS forecast for the monthly log exchange rate change for the country in the panel heading. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

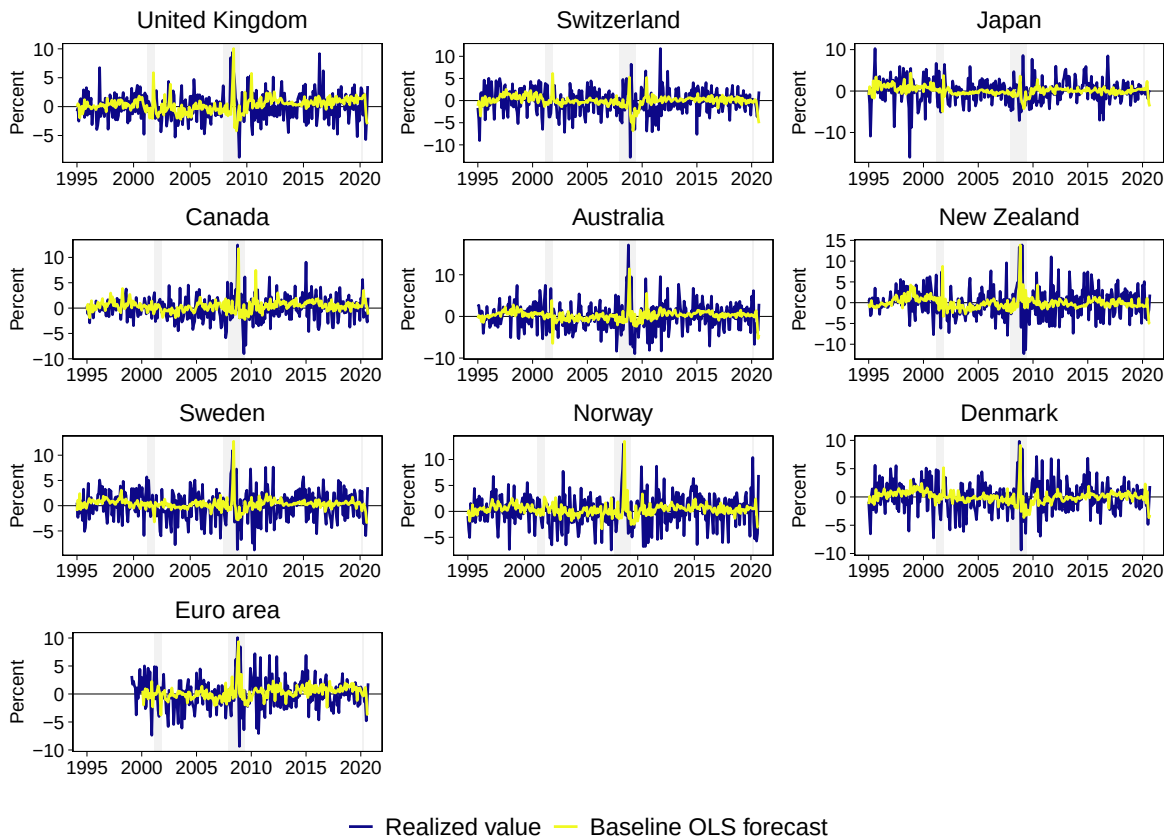


FIGURE IA.2

Baseline ENet Forecasts

Each panel depicts the realized value and baseline ENet forecast for the monthly log exchange rate change for the country in the panel heading. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

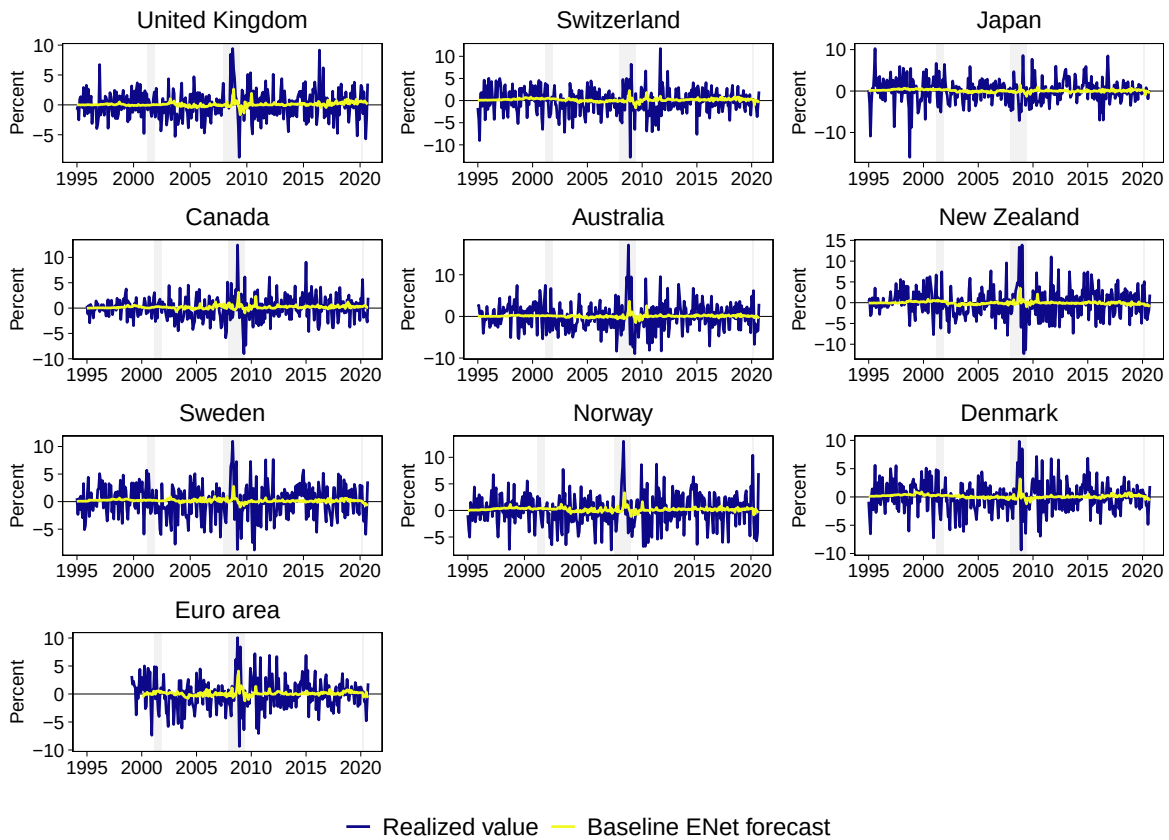


FIGURE IA.3

DNN Forecasts

Each panel depicts the realized value and DNN forecast for the monthly log exchange rate change for the country in the panel heading. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

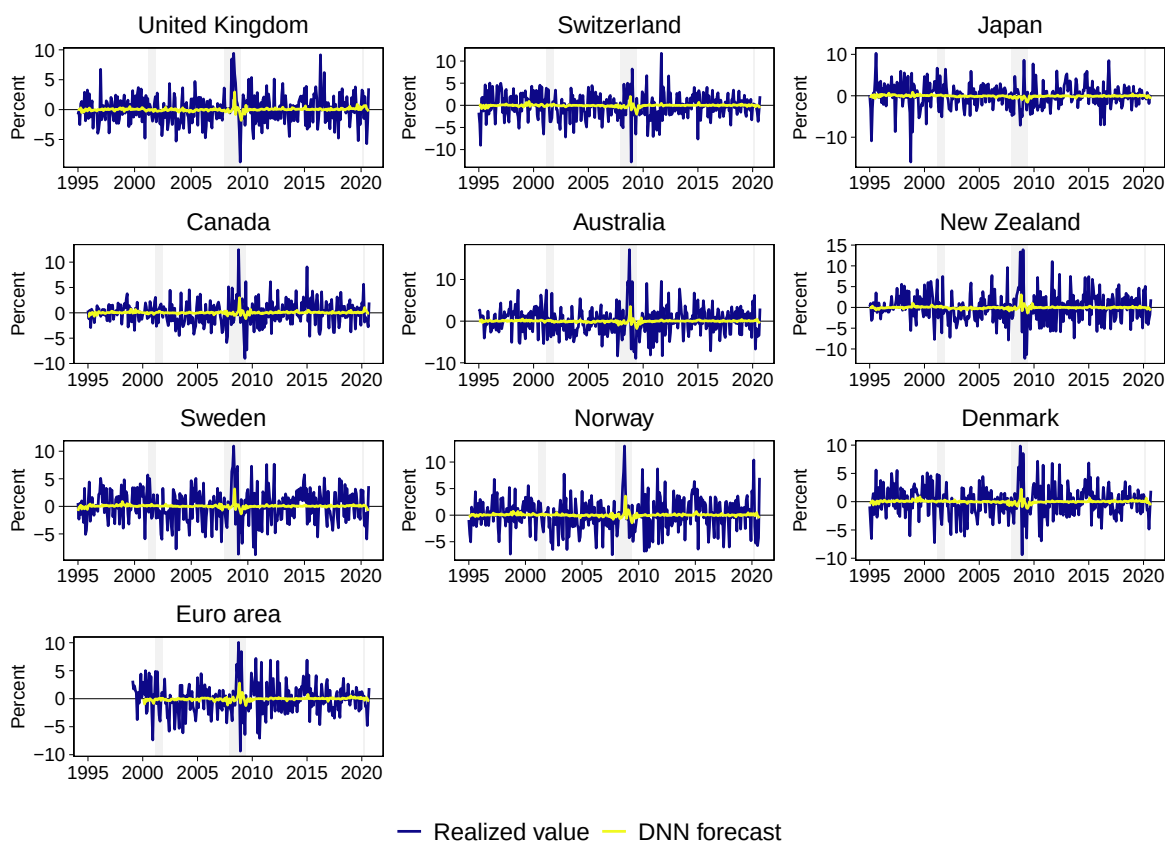


FIGURE IA.4

Ensemble Forecasts

Each panel depicts the realized value and ensemble forecast for the monthly log exchange rate change for the country in the panel heading. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

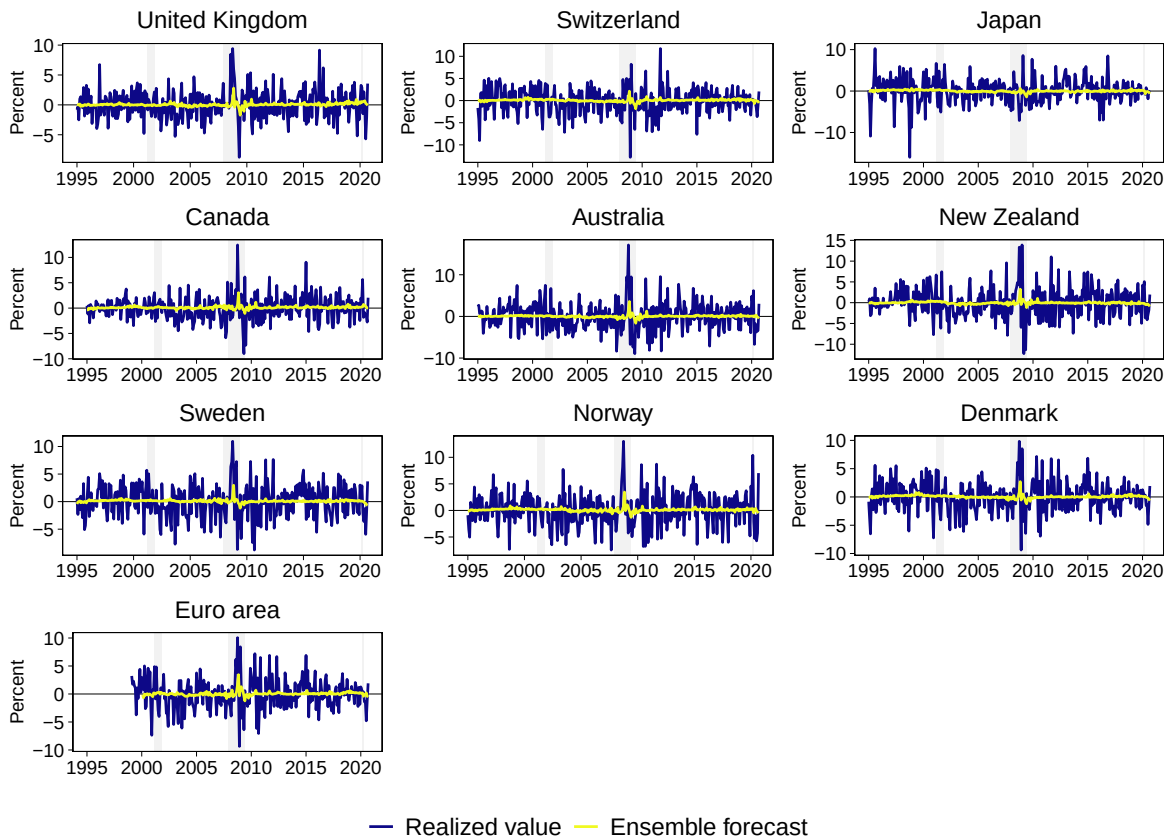


FIGURE IA.5

Rolling Out-of-Sample Clark-West Statistics

The figure depicts rolling 60-month Clark and West (2007) statistics for the baseline ENet, DNN, and ensemble forecasts. The horizontal axis gives the end of the 60-month period. The dotted horizontal line indicates the 10% critical value. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

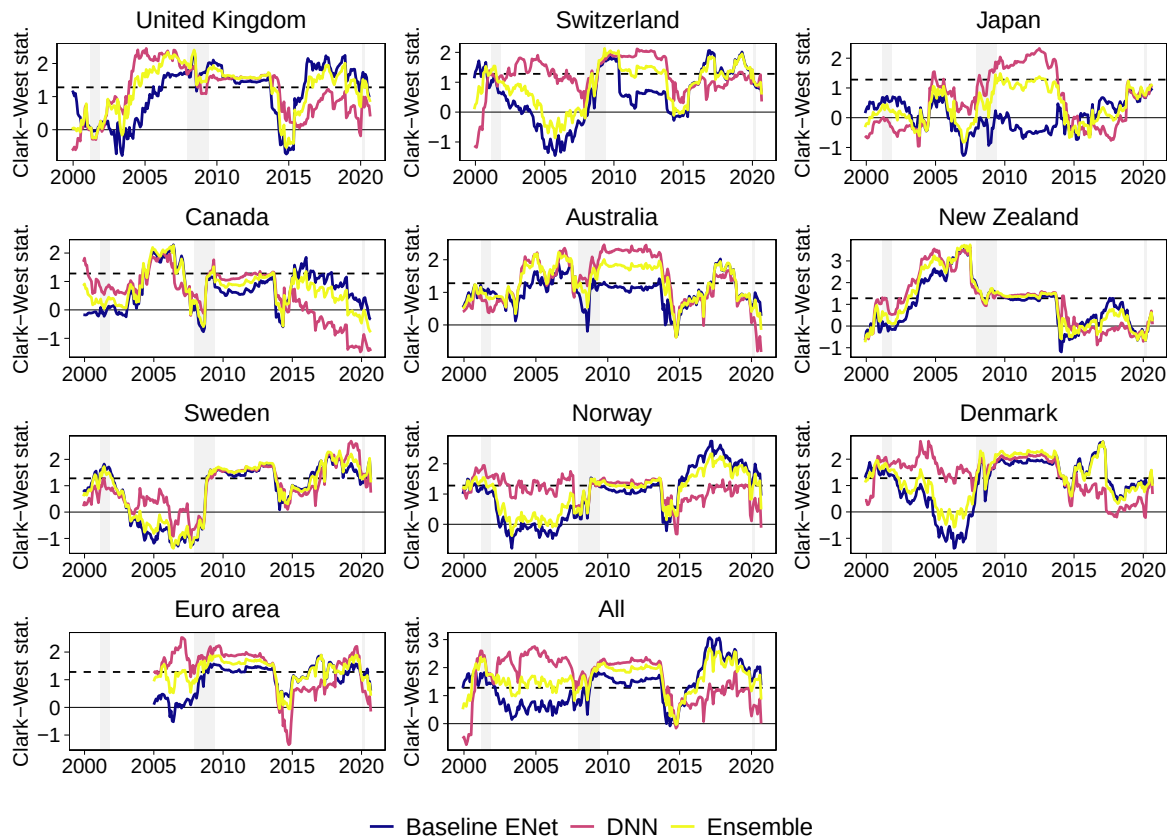


FIGURE IA.6

Cumulative Differences in Squared Prediction Errors

Each panel depicts the cumulative difference in squared prediction errors (CDSPE) for the country in the panel heading. The difference is computed for the no-change benchmark forecast relative to the baseline ENet, DNN, and ensemble forecasts (considered in turn). Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

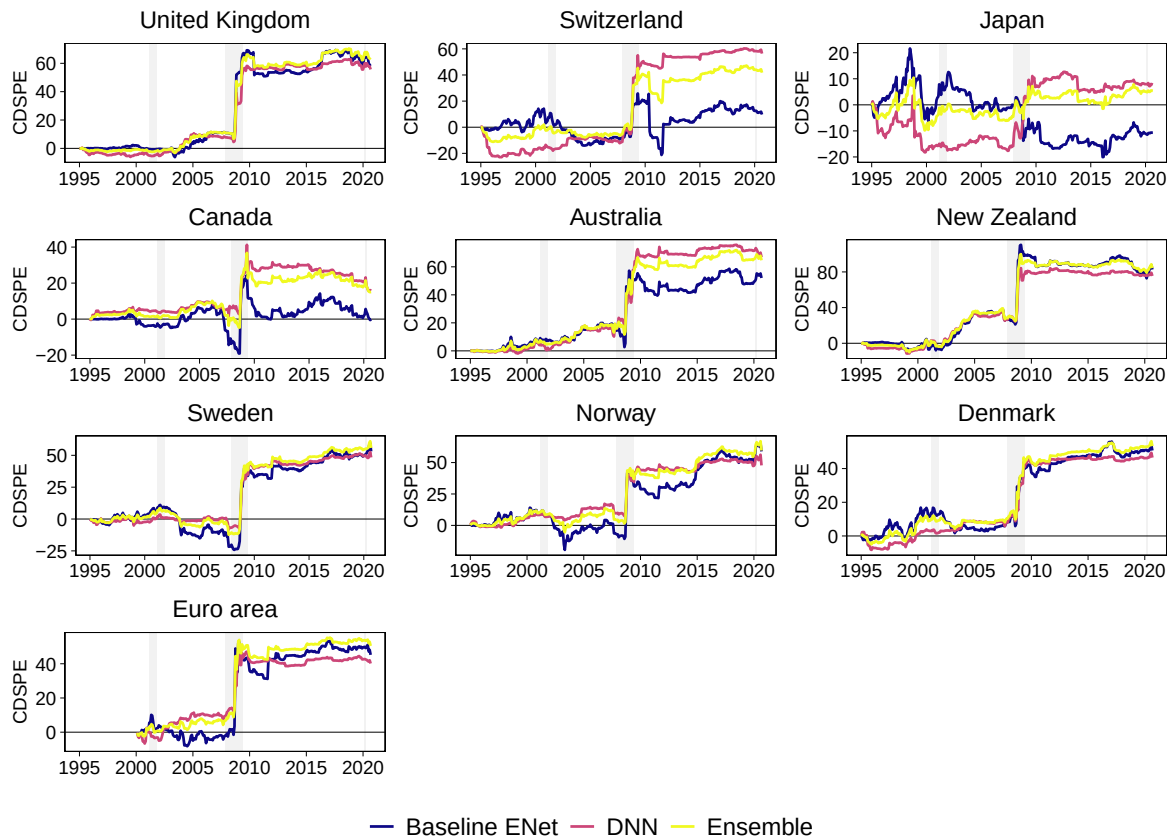


FIGURE IA.7

Change in US Foreign Bond Holdings

The figure shows actual and fitted values for a regression of the (standardized) change in US foreign bond holdings on the ten predictors selected by the ENet estimation of the baseline panel predictive regression (UN, DP, PE, INF.GVOL, UN.GVOL, BILL.GVOL, PE.EPU, PE.GILL, SRET12.GR, IV.GCOR). Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

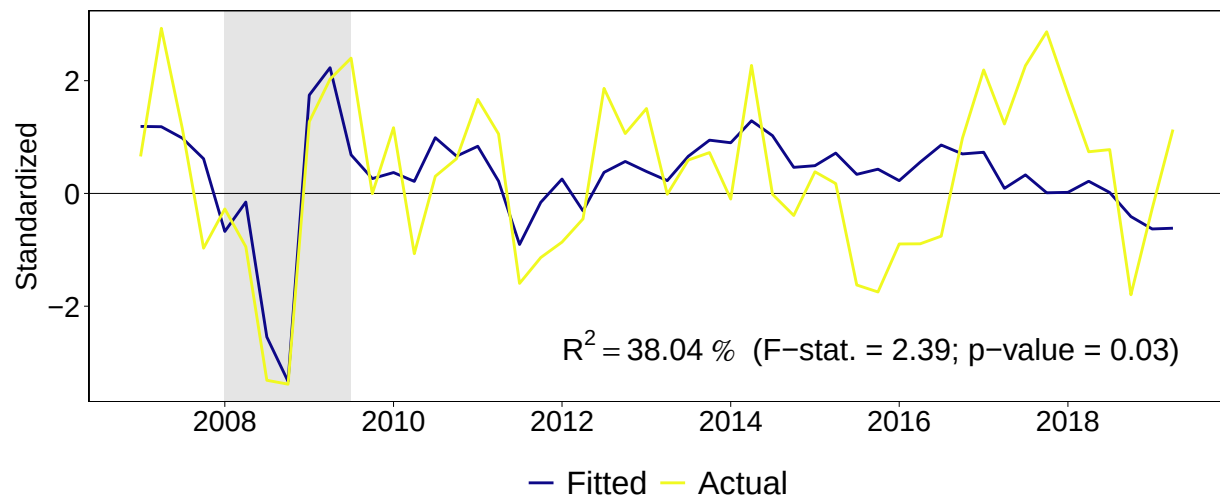


FIGURE IA.8

Full-Sample OLS Coefficient Estimates

The figure depicts OLS estimates of the slope coefficients for the baseline panel predictive regression and the 1985:01 to 2020:09 sample period. The predictors are standardized before estimation.

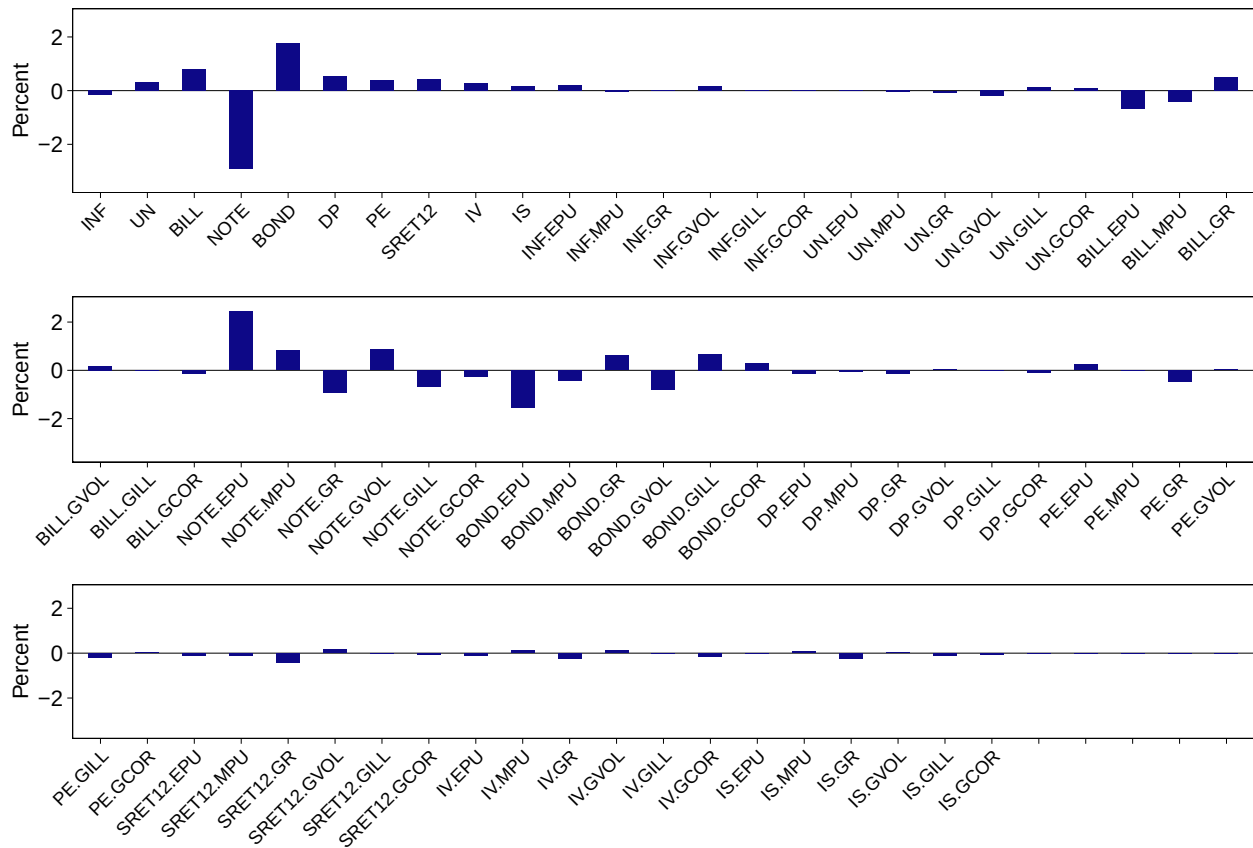


FIGURE IA.9

Recursive Baseline ENet Coefficient Estimates

Each panel depicts recursive ENet estimates of the slope coefficients for the baseline panel predictive regression for the predictor in the panel heading. Each point corresponds to the fitted model used to generate the forecast for the date on the horizontal axis. The predictors are standardized before estimation based on data available at the time of forecast formation.

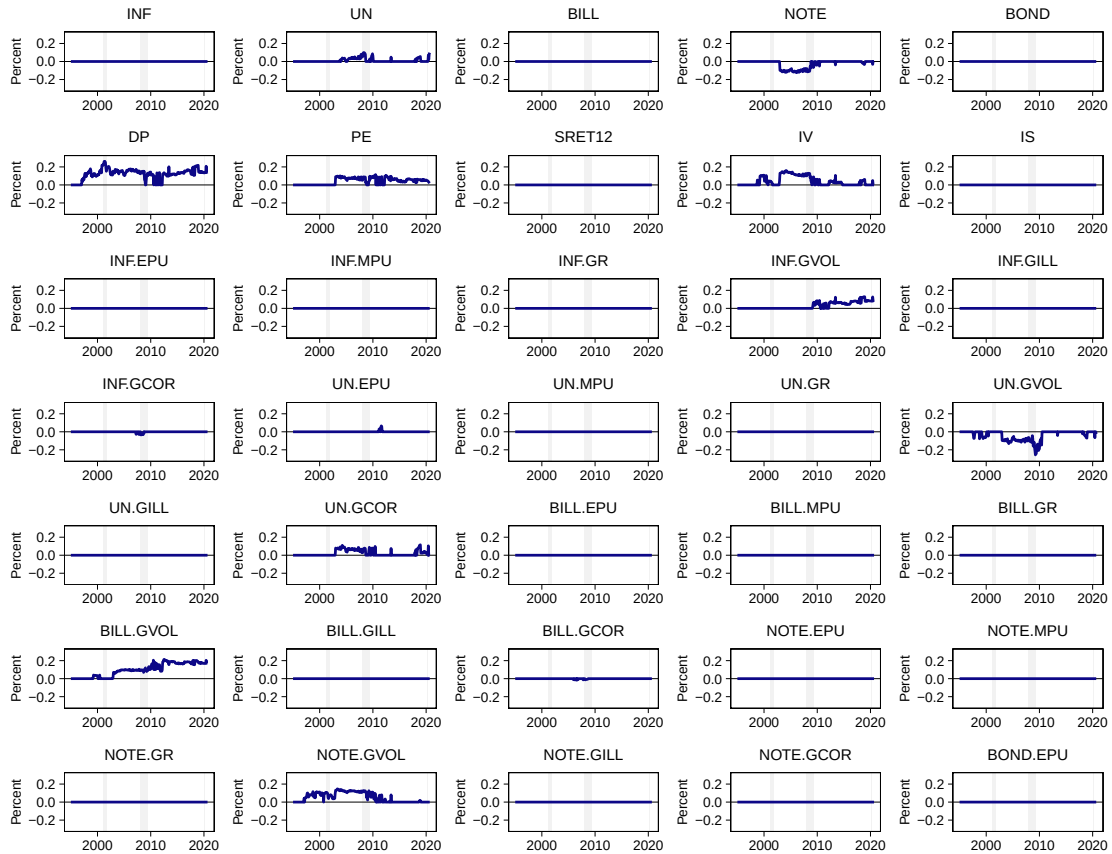


FIGURE IA.9 (continued)

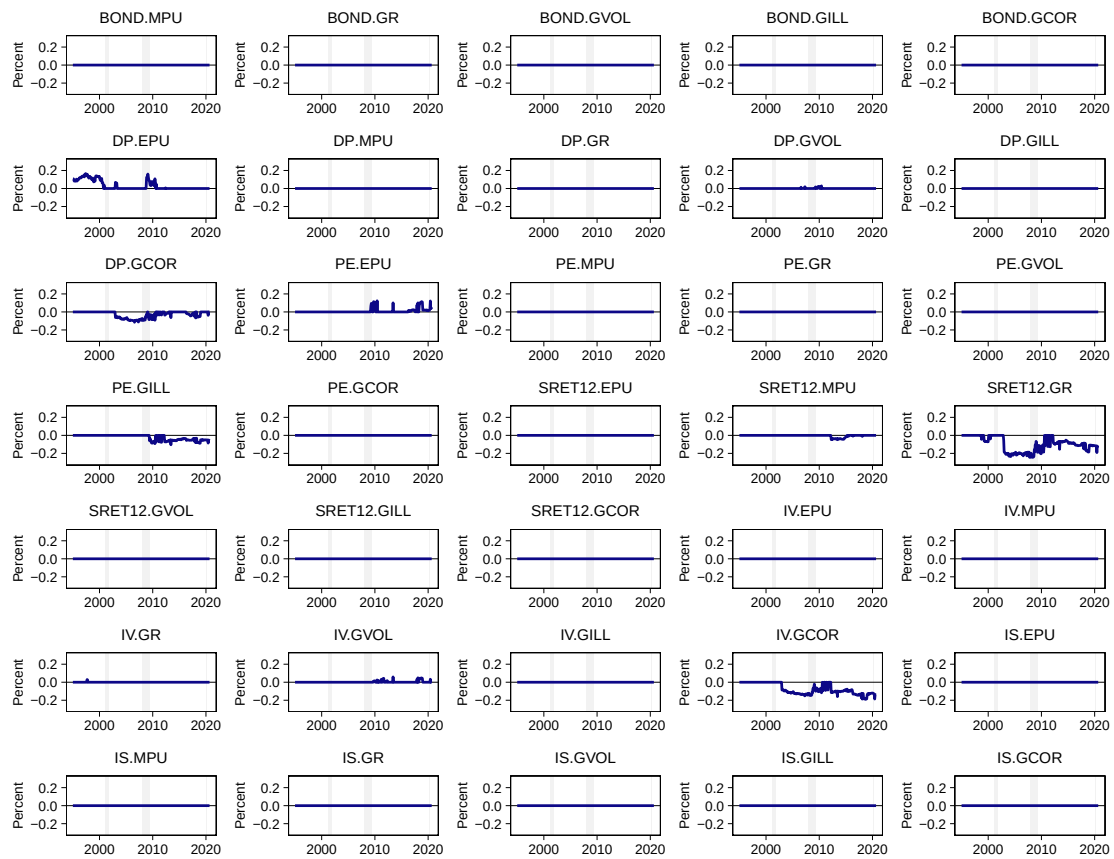


FIGURE IA.10

Recursive Variable Importance: DNN

Each panel depicts recursive PDP-based variable-importance measures for fitted DNN models for the predictor in the panel heading. Each point corresponds to the fitted model used to generate the forecast for the date on the horizontal axis.

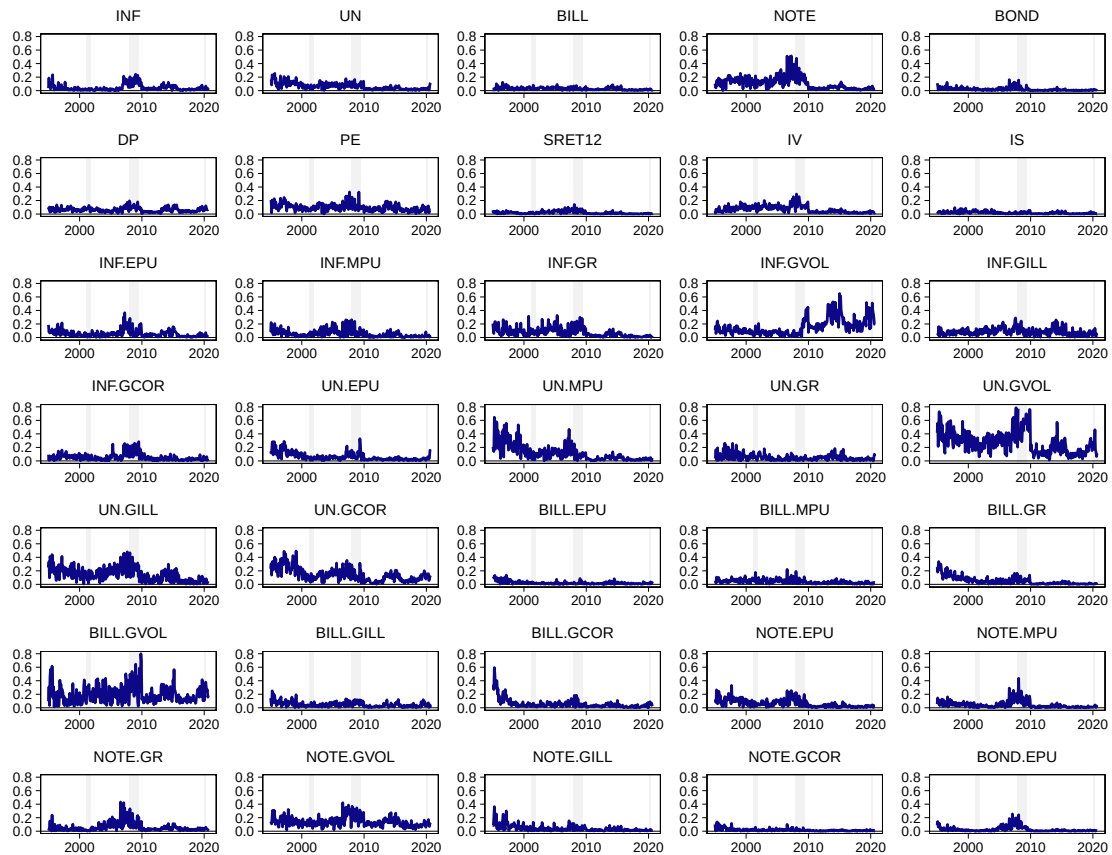


FIGURE IA.10 (continued)

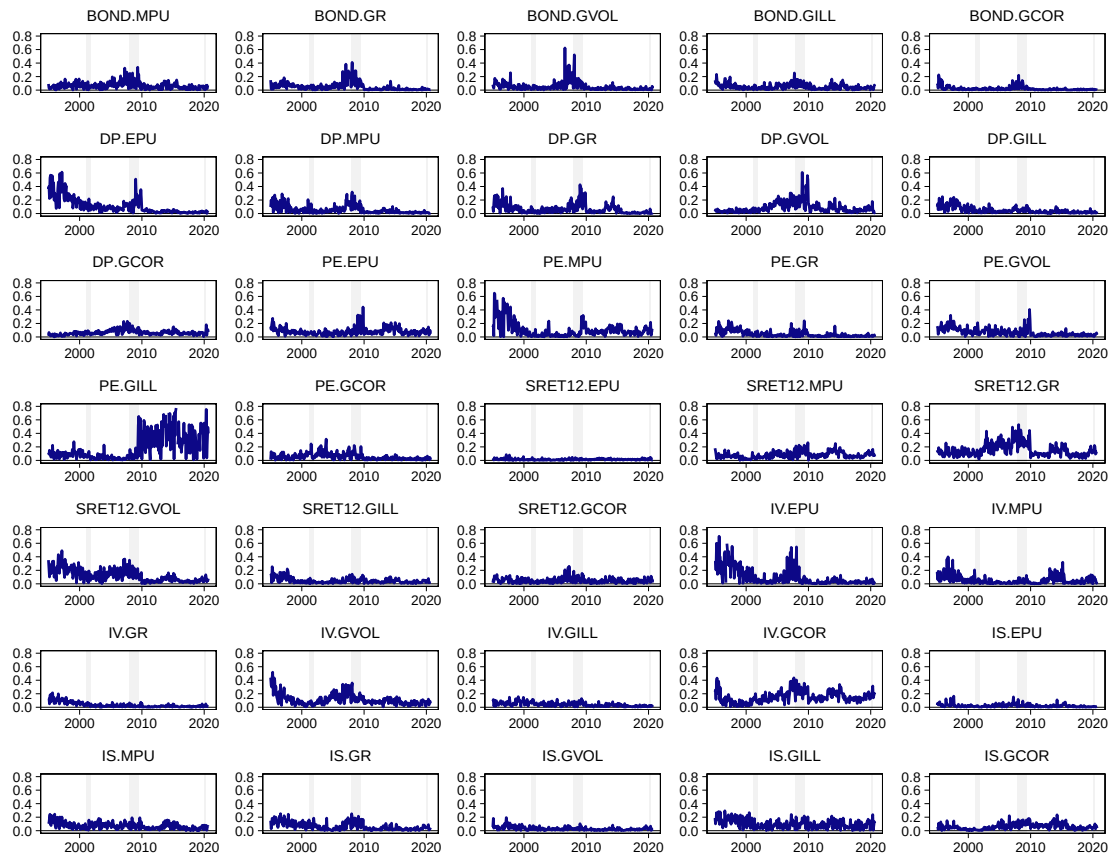


FIGURE IA.11

Recursive Variable Importance: Ensemble

Each panel depicts recursive PDP-based variable-importance measures for models underlying the ensemble forecast for the predictor in the panel heading. Each point corresponds to the fitted models used to generate the forecast for the date on the horizontal axis.

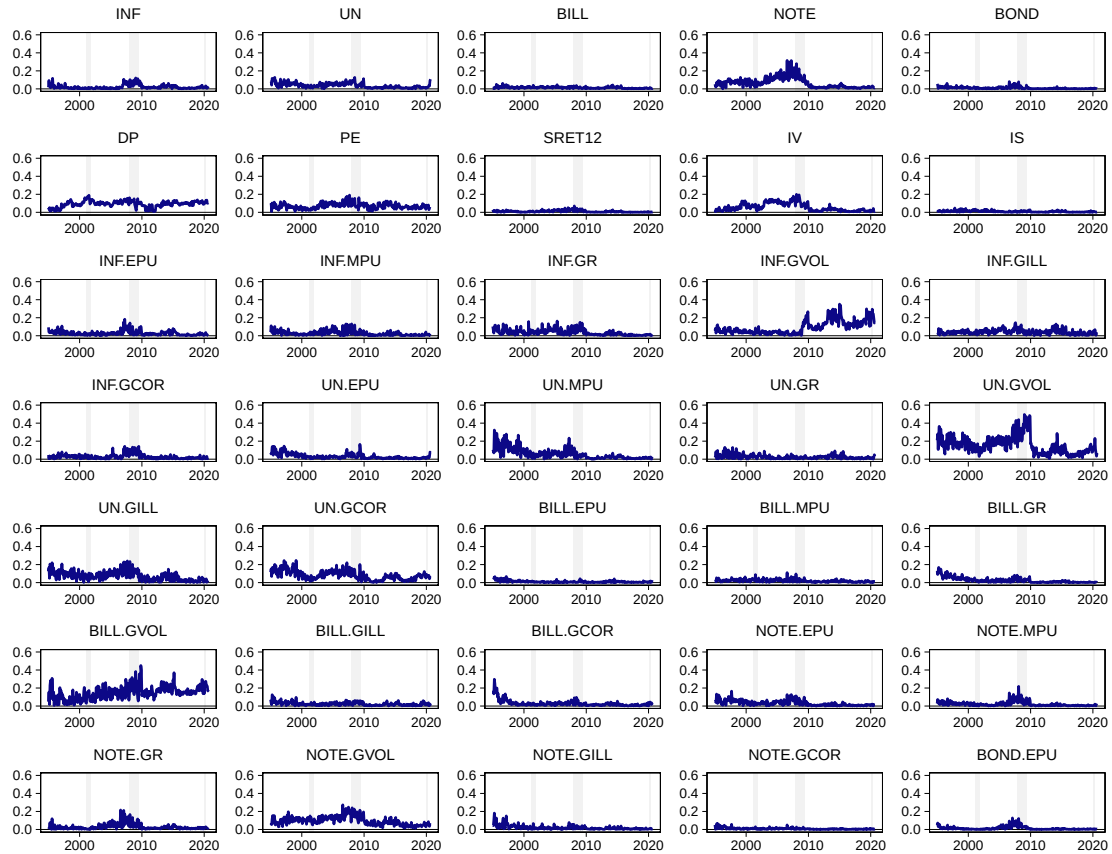


FIGURE IA.11 (continued)

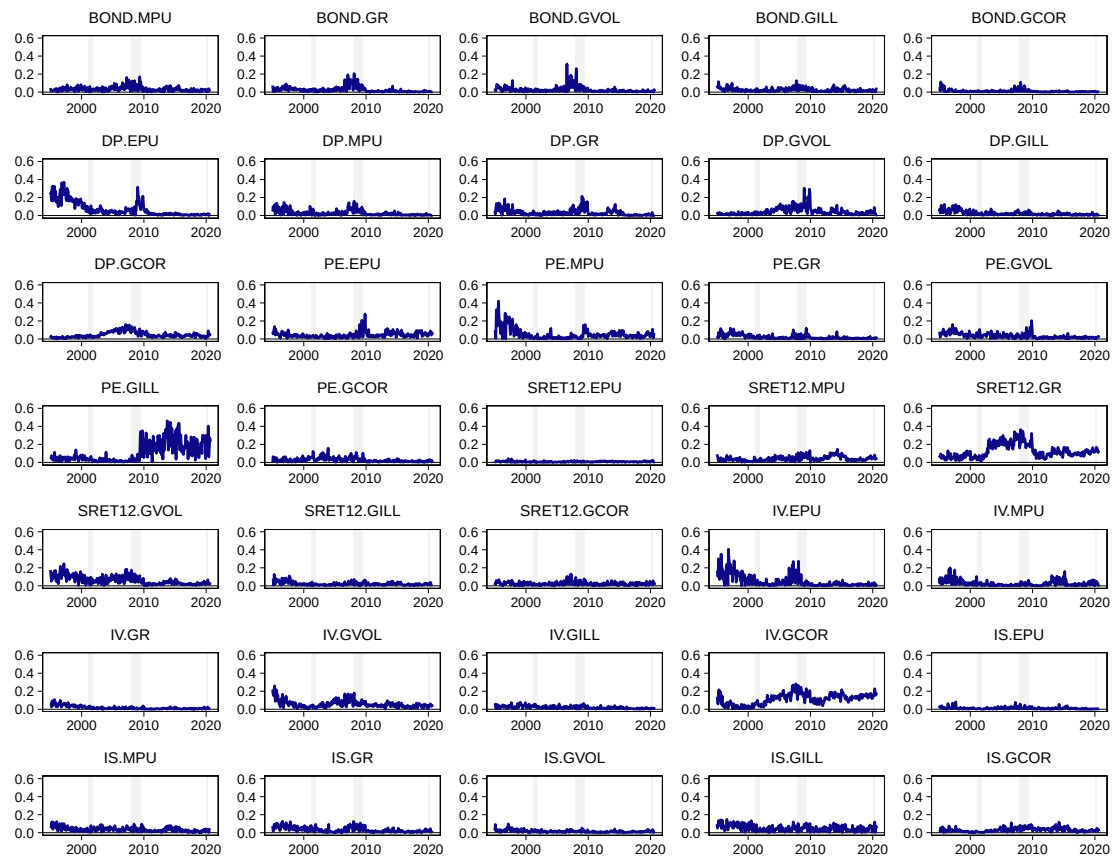


FIGURE IA.12

ML-Opt Portfolio Weights: Baseline ENet Forecasts

Each panel depicts weights for the country in the panel heading for the ML-Opt portfolio based on the baseline ENet forecast. The figure also includes weights for the Bench-Opt portfolio.

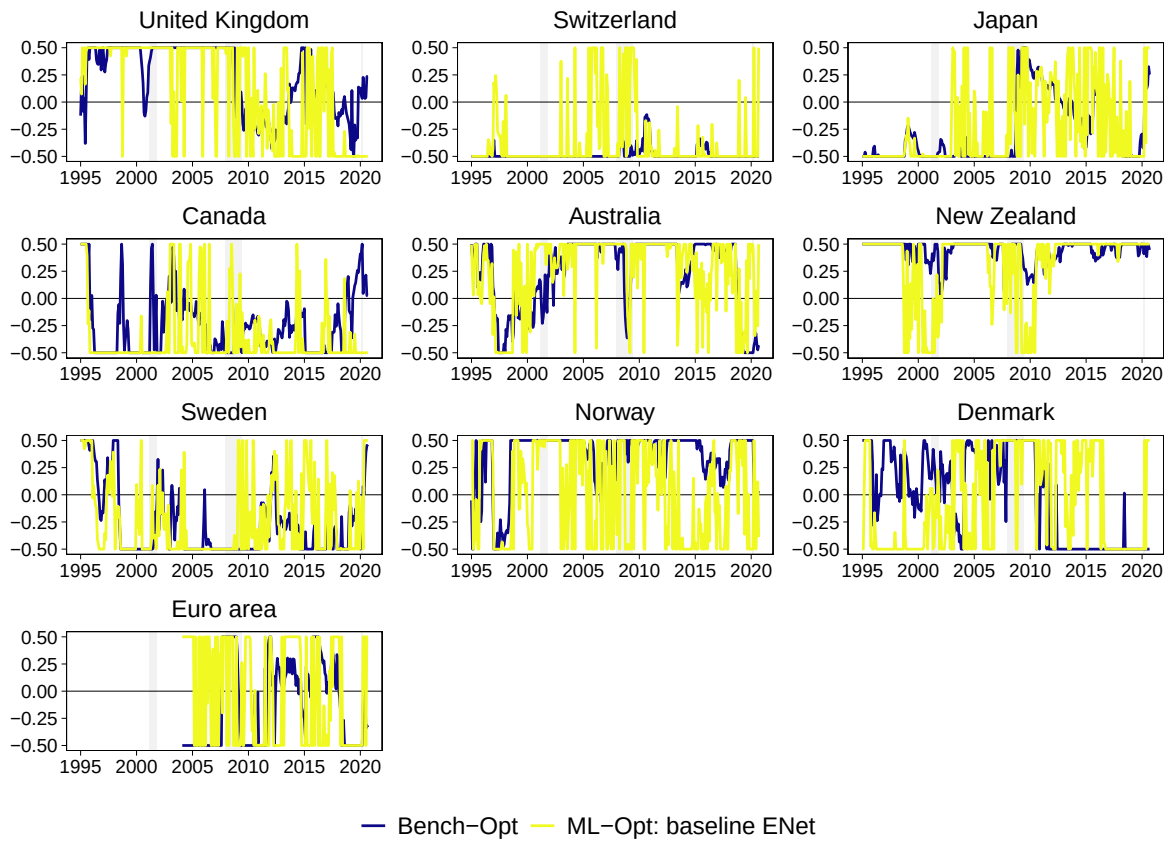


FIGURE IA.13

ML-Opt Portfolio Weights: DNN Forecasts

Each panel depicts weights for the country in the panel heading for the ML-Opt portfolio based on the DNN forecast. The figure also includes weights for the Bench-Opt portfolio.

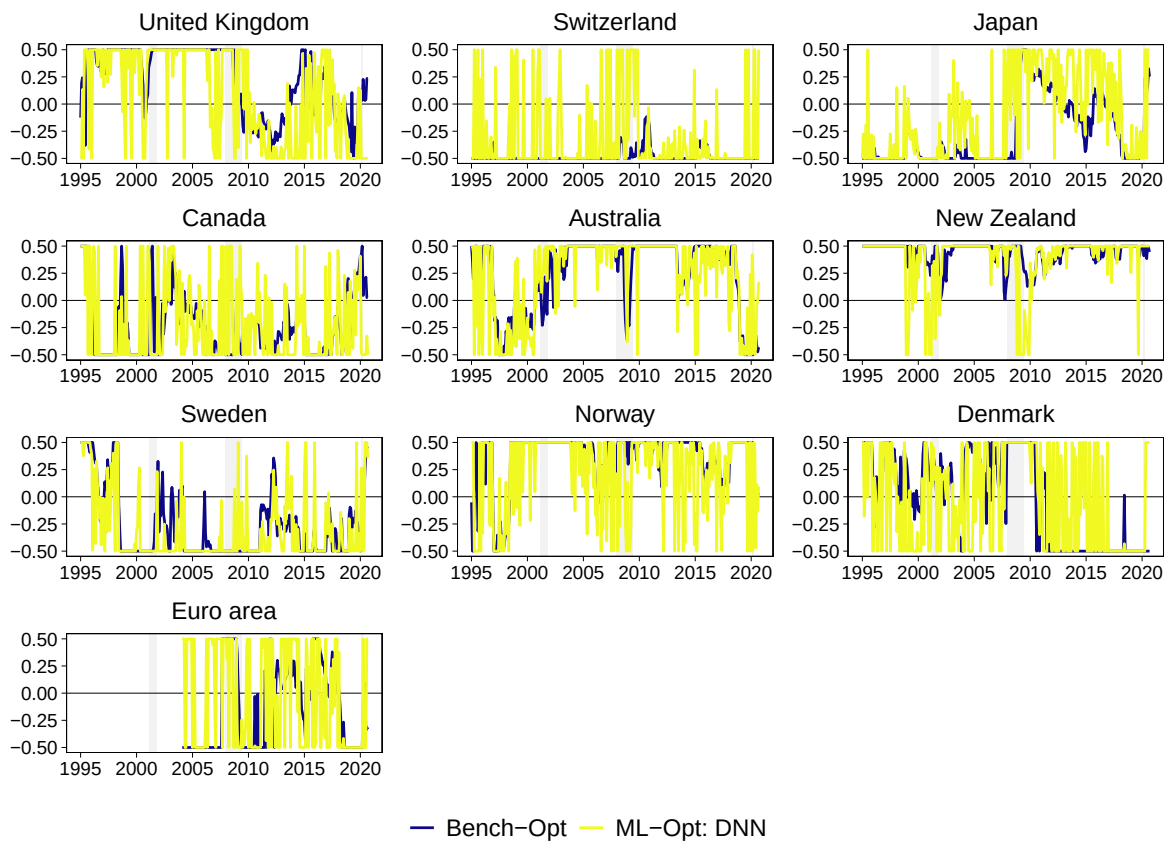


FIGURE IA.14

ML-Opt Portfolio Weights: Ensemble Forecasts

Each panel depicts weights for the country in the panel heading for the ML-Opt portfolio based on the ensemble forecast. The figure also includes weights for the Bench-Opt portfolio.

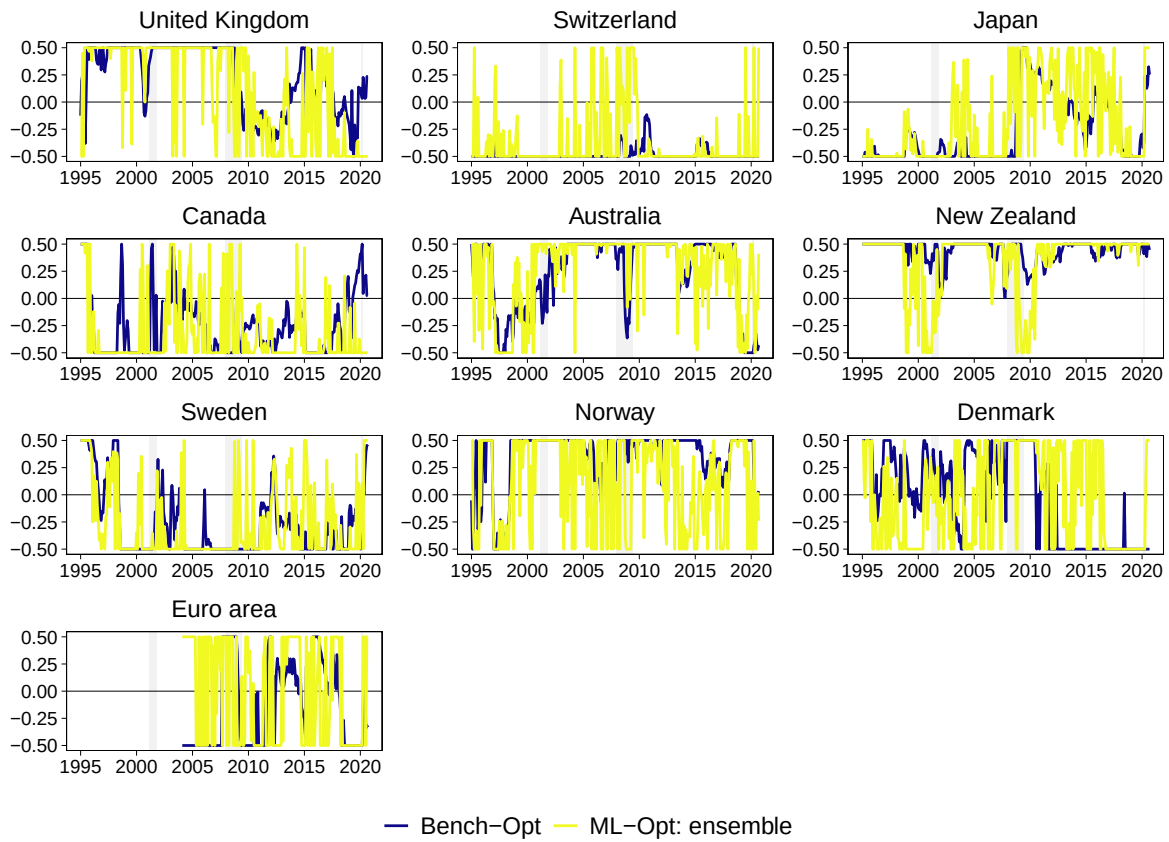


FIGURE IA.15

Currency Excess Return Forecasts in Late 2008: Baseline ENet

Each panel depicts currency excess return forecasts for the baseline ENet forecast of the exchange rate change, which serve as inputs for the ML-Opt portfolio, for the available currencies and month in the panel heading. Each panel also depicts currency excess return forecasts for the no-change benchmark forecast of the exchange rate change, which serve as inputs for the Bench-Opt portfolio, and realized currency excess returns.

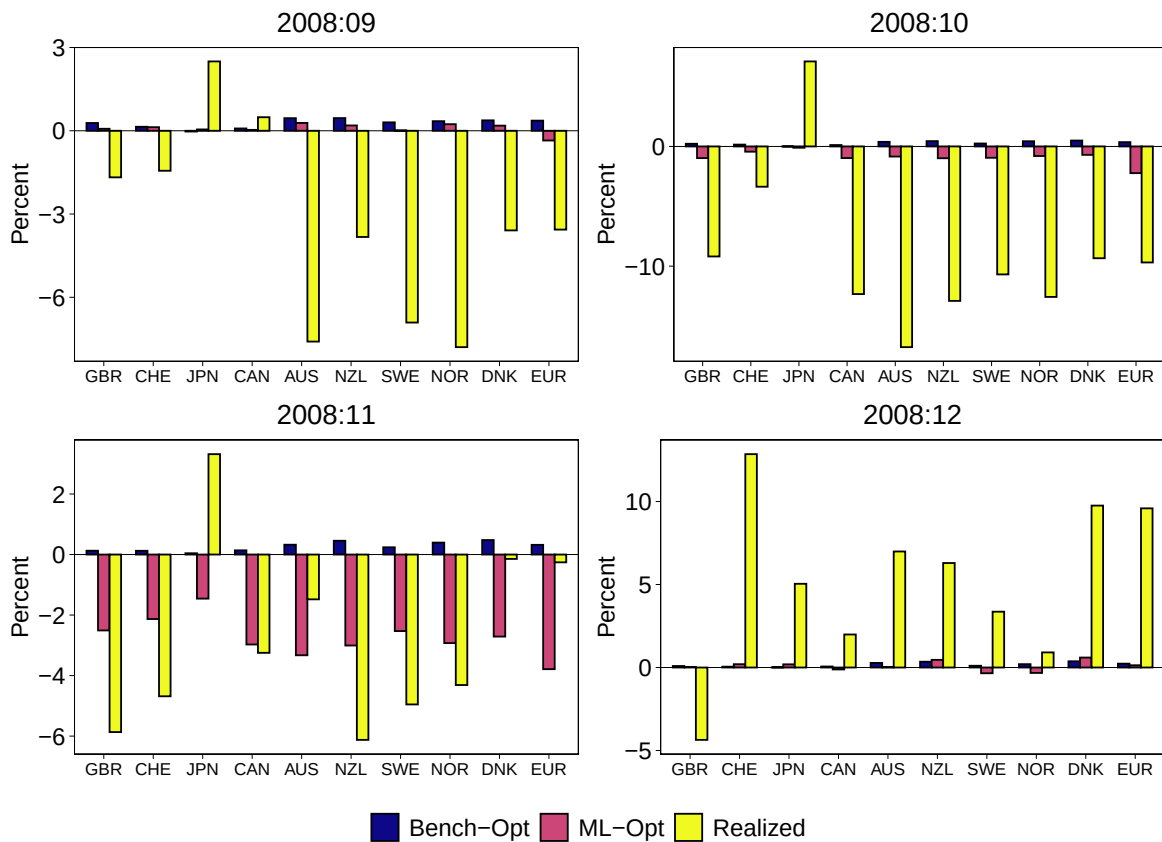


FIGURE IA.16

Currency Excess Return Forecasts in Late 2008: DNN

Each panel depicts currency excess return forecasts for the DNN forecast of the exchange rate change, which serve as inputs for the ML-Opt portfolio, for the available currencies and month in the panel heading. Each panel also depicts currency excess return forecasts for the no-change benchmark forecast of the exchange rate change, which serve as inputs for the Bench-Opt portfolio, and realized currency excess returns.

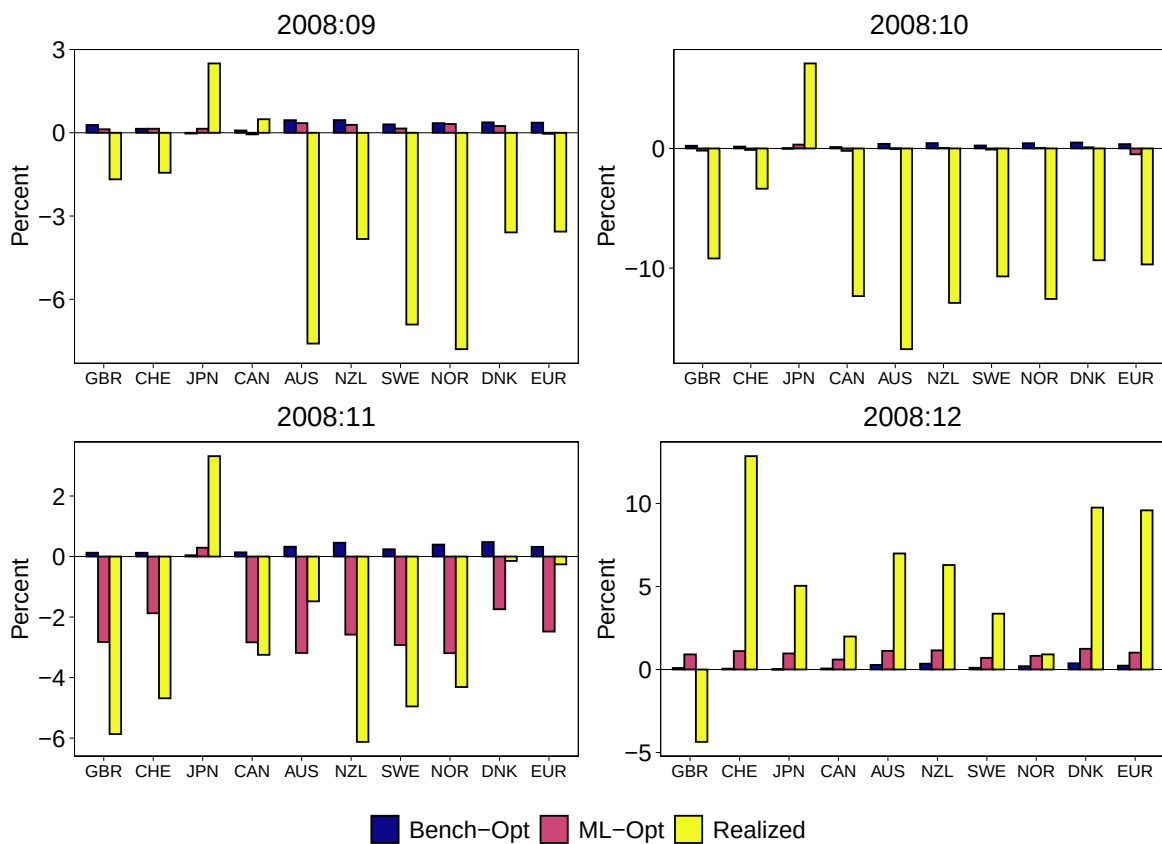


FIGURE IA.17

Currency Excess Return Forecasts in Late 2008: Ensemble

Each panel depicts currency excess return forecasts for the ensemble forecast of the exchange rate change, which serve as inputs for the ML-Opt portfolio, for the available currencies and month in the panel heading. Each panel also depicts currency excess return forecasts for the no-change benchmark forecast of the exchange rate change, which serve as inputs for the Bench-Opt portfolio, and realized currency excess returns.

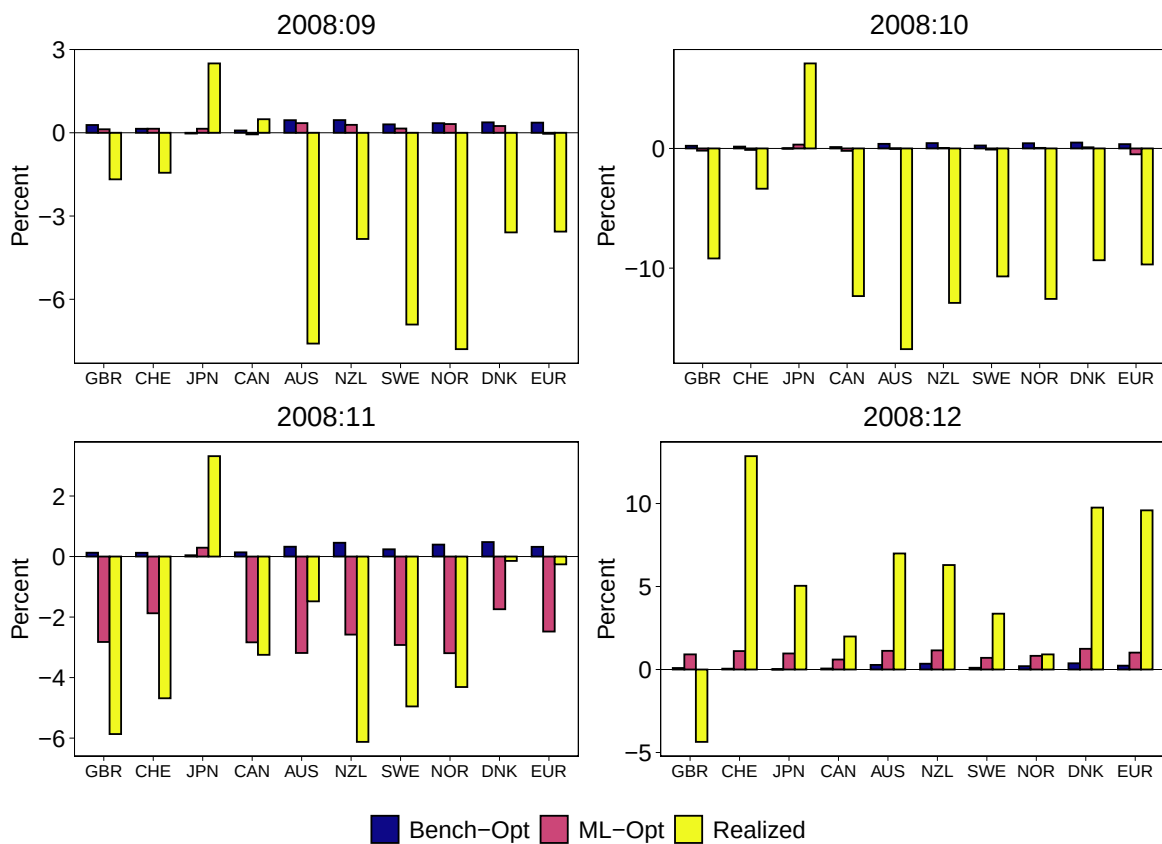


FIGURE IA.18

Portfolio Weights in Late 2008: Baseline ENet Forecast

Each panel depicts ML-Opt portfolio weights based on the baseline ENet forecast for the available currencies and month in the panel heading. Each panel also depicts Bench-Opt portfolio weights based on the no-change benchmark forecast.

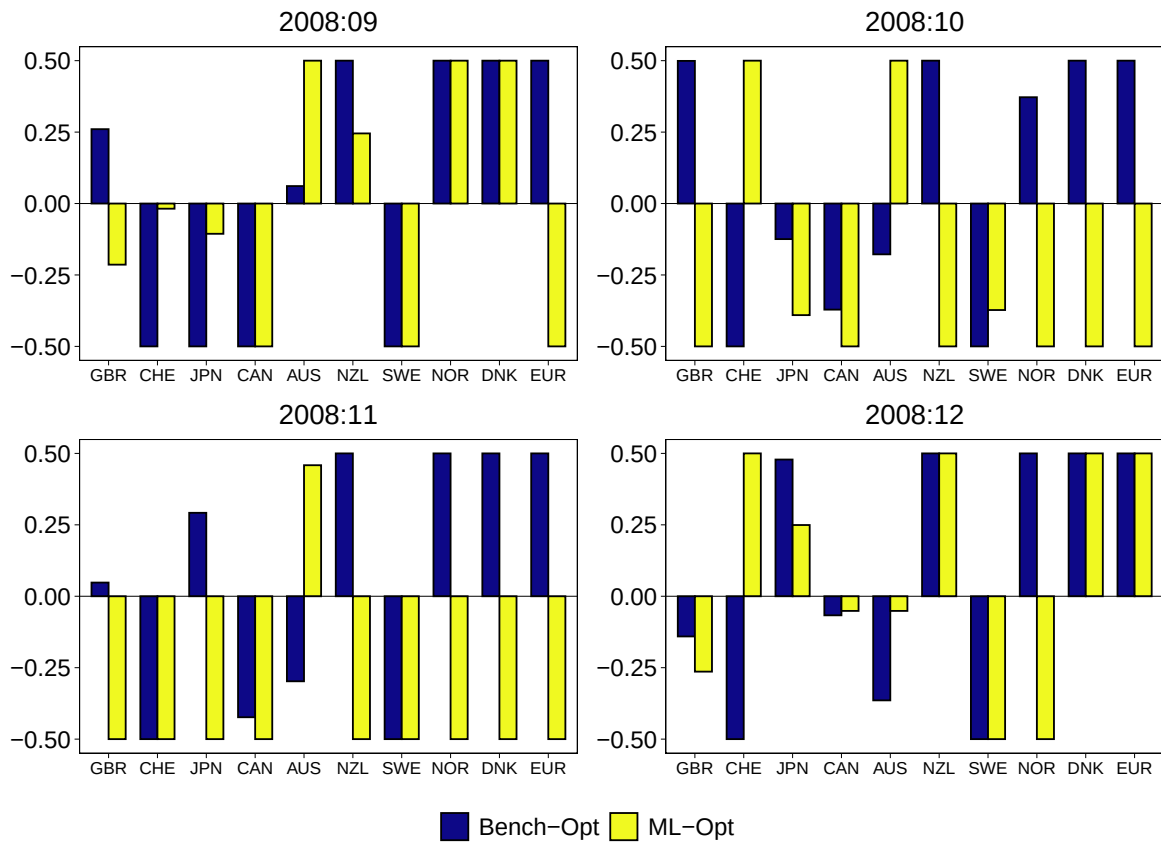


FIGURE IA.19

Portfolio Weights in Late 2008: DNN Forecast

Each panel depicts ML-Opt portfolio weights based on the DNN forecast for the available currencies and month in the panel heading. Each panel also depicts Bench-Opt portfolio weights based on the no-change benchmark forecast.

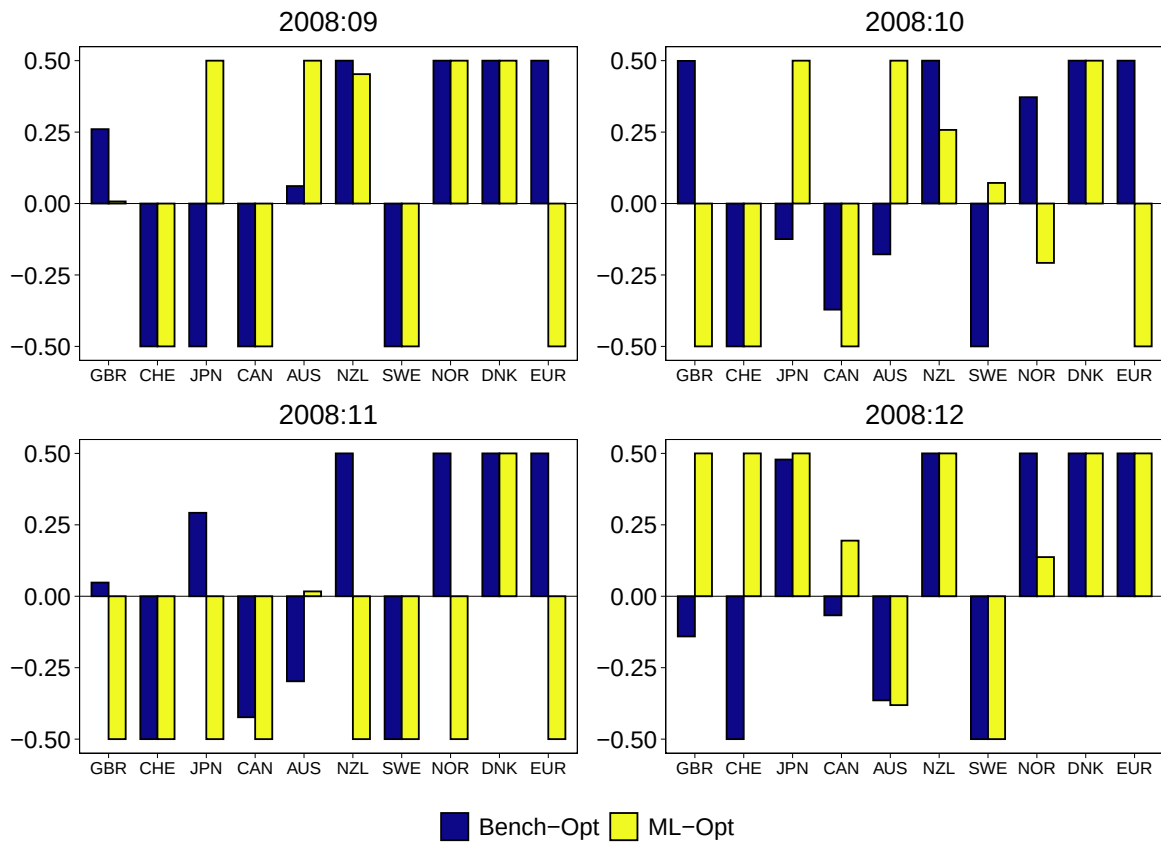


FIGURE IA.20

Portfolio Weights in Late 2008: Ensemble Forecast

Each panel depicts ML-Opt portfolio weights based on the ensemble forecast for the available currencies and month in the panel heading. Each panel also depicts Bench-Opt portfolio weights based on the no-change benchmark forecast.

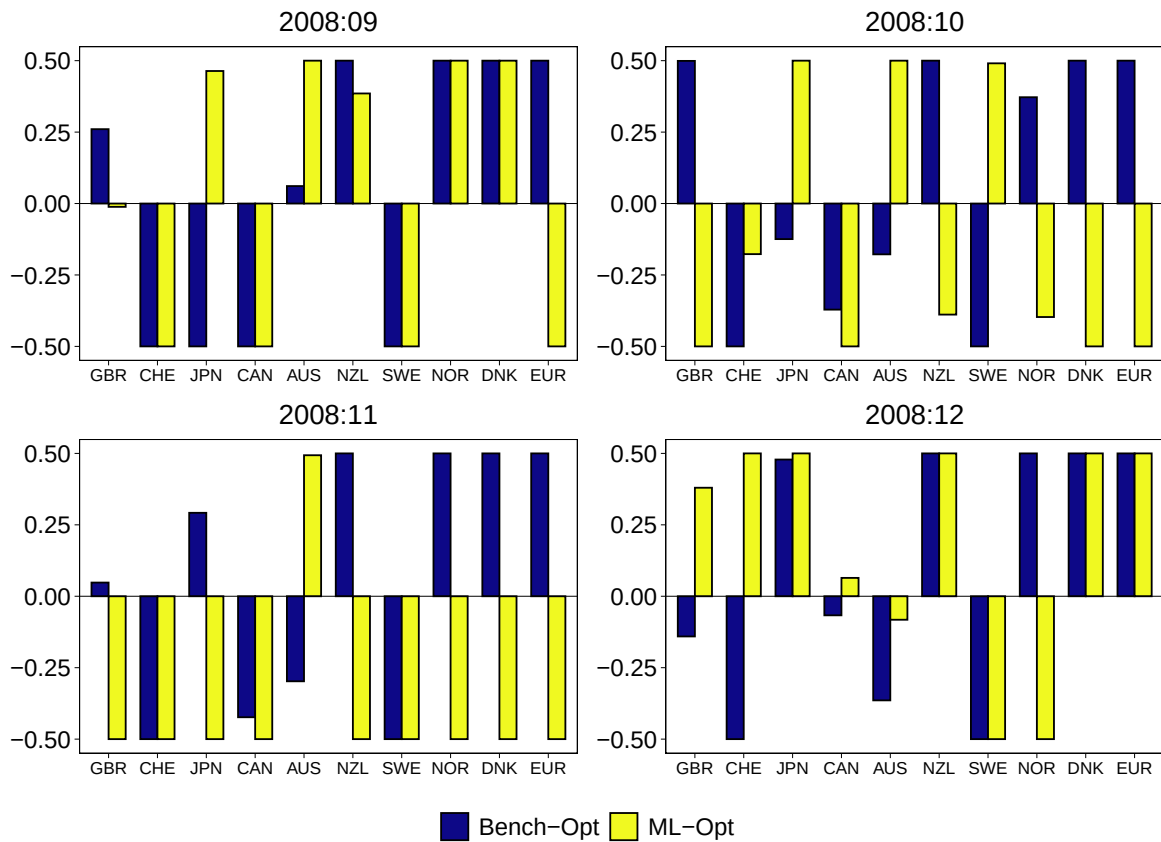


FIGURE IA.21

Conditional Maximum Sharpe Ratios

The figure depicts conditional maximum Sharpe ratios for currency excess return forecasts for the baseline ENet, DNN, ensemble, and no-change benchmark forecasts of the exchange rate change (considered in turn).

